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Project Title: - Digital Land and Property Management System in Dire Dawa.

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This is to certify that the final year project entitled:

"DIGITAL LAND AND PROPERTY MANAGEMENT SYSTEM"

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ABRIVATION AND ACRONOMYS

Acronym	Full Form
DLPMS	Digital Land and Property Management System
GIS	Geographic Information System
ICT	Information and Communication Technology
SQL	Structured Query Language
UI	User Interface
DBMS	Database Management System
LAN	Local Area Network
RAD	Rapid Application Development
GPS	Global Positioning System
API	Application Programming Interface
AWS	Amazon Web Services
CRUD	Create, Read, Update, and Delete
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
OS	Operating System

Acronym	Full Form
PDF	Portable Document Format
PNG	Portable Network Graphics
REST	Representational State Transfer
UML	Unified Modeling Language
ERD	Entity-Relationship Diagram
ER	Entity Relationship
CIC	Citizen Information Center
DDLA	Dire Dawa Land Administration
ETB	Ethiopian Birr
ET	Ethiopia
POSTGIS	PostgreSQL Geographic Information System Extension
RPC	Remote Procedure Call
SDLC	Software Development Life Cycle
UX	User Experience
RDBMS	Relational Database Management System
XML	Extensible Markup Language
VPN	Virtual Private Network

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Project

Land is one of the most valuable natural resources and a fundamental asset for any nation's economic development. In Ethiopia, land administration has historically been characterized by manual, paper-based systems that are prone to inefficiencies, errors, and corruption. The traditional methods of land and property management have resulted in numerous challenges including land disputes, unclear ownership rights, duplicate land holdings, and difficulties in accessing land information. These issues have significant implications for economic development, investment attraction, and social stability in the country.

The Federal Democratic Republic of Ethiopia has recognized the importance of modernizing land administration systems as part of its broader development agenda. The government has initiated various reforms aimed at improving land governance, including the establishment of land administration institutions at federal and regional levels. However, the transition from manual to digital systems has been gradual and uneven across different regions of the country. Many land administration offices still rely on outdated paper records that are vulnerable to damage, loss, and manipulation.

Digital transformation in land administration offers tremendous opportunities to address the existing challenges and improve service delivery. A Digital Land and Property Management System (DLPMS) can streamline processes, enhance transparency, reduce transaction costs, and provide secure storage and easy retrieval of land information. Such systems have been successfully implemented in various countries around the world, demonstrating significant improvements in land governance and economic outcomes. The proposed project aims to design and develop a comprehensive digital solution tailored to the Ethiopian context, taking into account the unique legal, institutional, and technological landscape of the country.

1.2 Statement of the Problem

The current land and property management system in Ethiopia faces numerous challenges that hinder efficient service delivery and contribute to disputes and conflicts. The manual, paper-based system is characterized by inefficiencies, including long processing times for land transactions, difficulties in retrieving historical records, and high costs associated with document verification. Citizens often spend considerable time and resources traveling to land administration offices multiple times to complete simple transactions, which imposes significant burdens on both individuals and businesses.

One of the most pressing issues is the prevalence of land disputes arising from unclear or conflicting ownership claims. The lack of a centralized, digital database makes it difficult to verify the authenticity of land titles and detect duplicate registrations. This situation has led to numerous cases where multiple individuals claim ownership of the same parcel of land, resulting in lengthy legal battles and social conflicts. Furthermore, the vulnerability of paper records to damage from fire, water, pests, and general deterioration poses significant risks to the integrity of land information.

The absence of a digital system also creates opportunities for corrupt practices, as manual records can be altered or destroyed without leaving traces. Land administration officials may face pressure to manipulate records for personal gain, undermining public trust in land governance institutions. Additionally, the current system lacks integration with other government services, requiring citizens to navigate multiple offices and procedures to complete land-related transactions. These problems collectively impede economic development, discourage investment, and perpetuate poverty in both urban and rural areas of Ethiopia.

1.2 Objectives of the Project

1.2.1 General Objective

To develop a digital land and property management system for Dire Dawa City.

1.2.2 Specific Objectives

The specific objectives of this project are:

- ✓ To analyze the current land management system in Dire Dawa.
- ✓ To design web and mobile user interfaces.
- ✓ To develop a centralized and secure database.
- ✓ To implement GIS for land mapping and boundary verification.
- ✓ To develop a citizen service portal.
- ✓ To develop an administrative management dashboard.
- ✓ To implement role-based authentication and audit logging.
- ✓ To test system usability, performance, and security.

1.3 Scope and Limitation of the Project

The scope of this project encompasses the design and development of a web and mobile based Digital Land and Property Management System tailored specifically to Dire Dawa City. The system will cover core functionalities including land registration, property transfer, record management, search and retrieval, reporting, and user management. The geographical focus will be on urban land parcels within Dire Dawa City limits, specifically targeting selected kebeles as a pilot implementation area, with the potential for future expansion to other areas. The system will integrate with Geographic Information System (GIS) technology for spatial data management, enabling the visualization and analysis of land parcels. It will support multiple user roles including administrators, land officers, tax officials, and citizens, each with appropriate access permissions. The development will prioritize compliance with Ethiopian land laws and regulations, including the Urban Land Lease Holding Proclamation and Dire Dawa City land policies.

However, certain limitations must be acknowledged. The project will not cover the digitization of existing paper records, which would require a separate data migration initiative. The system will be developed as a prototype with core functionalities for selected kebeles, and full city-wide deployment would require additional resources and institutional arrangements. Real payment gateway integration will be simulated due to regulatory and banking constraints. Furthermore, the project will focus on technical development aspects and may not address all organizational and capacity-building needs for successful implementation.

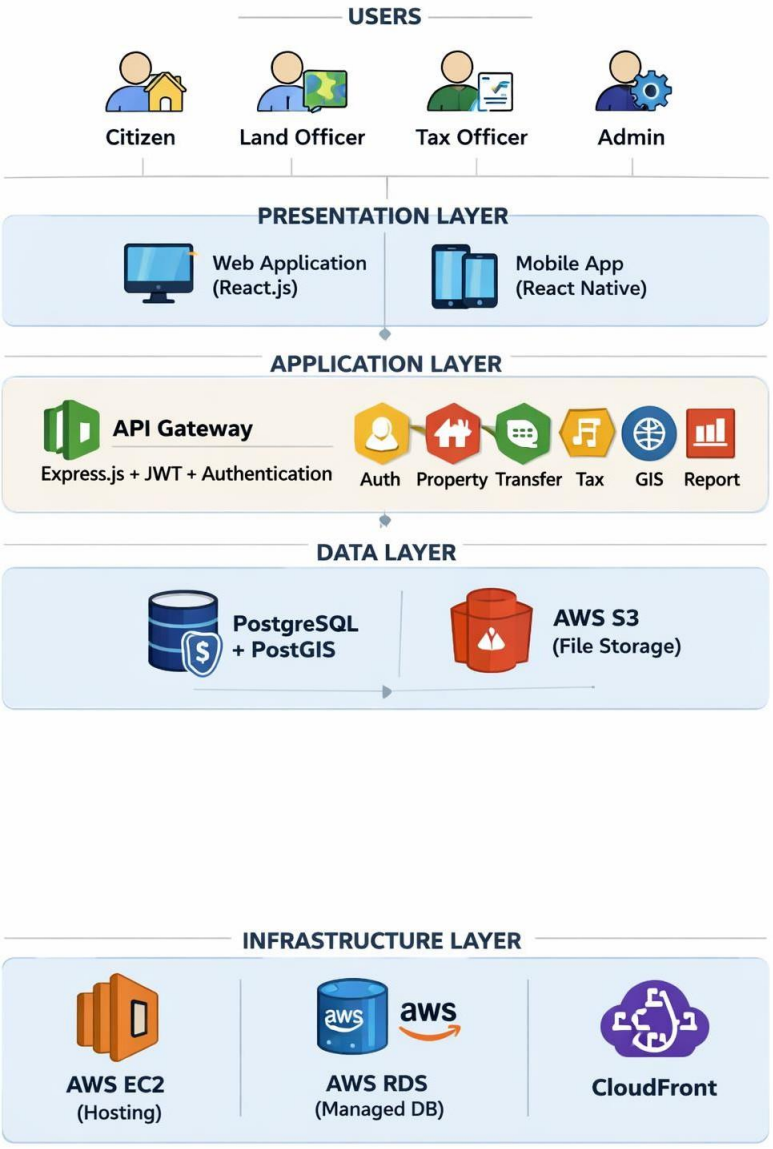


Figure 1 System Architecture

1.4 Significance of the Project

The implementation of a Digital Land and Property Management System will have far-reaching significance for Ethiopia's development agenda. From an economic perspective, secure and transparent land rights are essential for investment attraction, agricultural productivity, and urban development. The system will reduce transaction costs and processing times, enabling more efficient land markets and stimulating economic activity. Clear land titles also enable landowners to use their property as collateral for credit, expanding access to finance for both individuals and businesses.

land administration. Digital records with audit trails will deter corrupt practices and provide evidence for investigating any irregularities. The system will also improve service delivery by enabling faster processing of land transactions, reducing the burden on both citizens and government officials. Citizens will be able to access information about their land records and track the status of their applications, increasing trust in land administration institutions.

The project will also contribute to dispute reduction and social stability by providing clear and verifiable land records. With accurate and accessible information, many potential disputes can be prevented before they escalate. When disputes do arise, the system will provide reliable evidence for resolution. Additionally, the project will generate valuable data for urban planning, infrastructure development, and natural resource management, supporting evidence-based decision-making by government authorities.

1.5 Beneficiaries of the Project

The Digital Land and Property Management System will benefit multiple stakeholders across Ethiopian society. Primary beneficiaries include individual landowners and property holders who will gain improved access to secure and verifiable land records. They will benefit from faster processing of land transactions, reduced transaction costs, and enhanced protection of their property rights. The system will also benefit prospective land buyers and investors who will be able to verify ownership and conduct due diligence more efficiently.Government institutions at federal, regional, and local levels will benefit from improved efficiency in land administration operations. Land administration offices will be able to serve more citizens with existing resources, reduce processing times, and minimize errors in record-keeping. The system will also support tax collection by providing accurate information about property ownership and values. Urban planning authorities will benefit from spatial data for infrastructure development and land use planning.

The judiciary and legal profession will benefit from access to reliable land records for dispute resolution. Banks and financial institutions will be able to verify collateral for mortgage loans with greater confidence. Development organizations and researchers will have access to aggregated data for policy analysis and program design. Ultimately, Ethiopian society as a whole will benefit from improved land governance, reduced conflicts, and enhanced economic opportunities.

1.6 Methodology

1.6.1 Data Gathering Methodology

The project will employ a multi-faceted approach to data gathering to ensure comprehensive understanding of requirements and context. Primary data will be collected through interviews with key stakeholders including land administration officials, legal experts, and potential system users. Structured questionnaires will be administered to citizens who interact with land administration services to understand their experiences and expectations. Direct observation of current land administration processes will provide insights into workflow inefficiencies and user needs.

Secondary data will be gathered through review of existing documents including land administration laws and regulations, government policy documents, academic literature on land management systems, and technical reports on similar implementations in other countries. Document analysis will help understand the legal framework governing land administration in Ethiopia and identify compliance requirements for the system. Comparative analysis of digital land management systems implemented in other African countries will provide lessons and best practices.

1.6.2 Development Methodology

The project will adopt the Rapid Application Development (RAD) methodology for system development. RAD is particularly suitable for this project due to its emphasis on rapid prototyping, iterative development, and active user involvement. The methodology enables quick delivery of functional components while allowing for continuous refinement based on user feedback. This approach is well-suited to the Ethiopian context where requirements may evolve as stakeholders gain better understanding of digital possibilities.

The development process will proceed through several phases including requirements planning, user design, construction, and implementation. The requirements planning phase will involve extensive stakeholder consultation to define system specifications. The user design phase will create prototypes that stakeholders can review and refine. The construction phase will involve actual coding and testing of the system. The implementation phase will deploy the system and provide training for users. Throughout these phases, continuous feedback loops will ensure that the system meets stakeholder need

1.6.3 Development Tools

The development of the Digital Land and Property Management System will utilize a modern technology stack selected for reliability, scalability, and accessibility. The following table presents the key development tools and technologies that will be employed:

Table 1 Methedology

Tier	Technology	Purpose
Web Frontend	React.js, Tailwind CSS, Redux	Desktop web interface with modern styling
Mobile Frontend	React Native, SQLite	Android/iOS apps with offline storage
Mobile Features	React Native Camera, Geolocation	GPS tracking and document scanning
Backend	Node.js, Express, JWT	REST API, authentication
Database	PostgreSQL 15, PostGIS	Data storage, spatial data management
DevOps	Git, GitHub, VS Code, Postman	Development, testing, version control
Cloud	AWS EC2, S3	Hosting, backup, document storage

1.7 Feasibility Study

1.7.1 Economic Feasibility

The economic feasibility of the Digital Land and Property Management System is highly favorable when considering both costs and benefits. The development costs include hardware procurement, software licenses, development personnel, and training expenses. While initial investment is required, the long-term economic benefits far outweigh these costs. The system will reduce operational costs by eliminating paper-based processes, reducing manual data entry, and streamlining workflows. Processing time reductions will free up staff time for other productive activities.

The system will generate economic benefits through improved revenue collection from land transactions and property taxes. More accurate records will enable better assessment of property values and ensure that appropriate fees and taxes are collected. The reduction in land disputes will save considerable resources currently spent on legal proceedings and conflict resolution. Furthermore, secure land rights will stimulate investment and economic activity, generating broader economic benefits for the Ethiopian economy.

1.7.2 Technical Feasibility

The technical feasibility of the project is supported by the availability of mature technologies and skilled personnel. The proposed technology stack using PHP/Laravel, MySQL, and open-source GIS tools is well-established and widely used for similar applications globally. These technologies have proven track records of reliability, security, and scalability. The development team possesses the necessary technical skills in web development, database design, and GIS implementation required for successful project completion.Ethiopia's growing ICT infrastructure, including expanding internet connectivity and power supply, provides a foundation for system deployment. The system can be designed to function with varying levels of connectivity, including offline capabilities for areas with limited internet access. Cloud hosting options are increasingly available and affordable, reducing the need for substantial hardware investments. Technical support and maintenance resources are available through local IT service providers and the broader open-source community.

1.7.3 Operational Feasibility

The operational feasibility of the system depends on organizational readiness and user acceptance. Land administration institutions in Ethiopia have been undergoing modernization efforts and have expressed interest in digital solutions. The system will be designed with

user-friendly interfaces that minimize training requirements and facilitate adoption. The proposed system aligns with government policies promoting e-governance and digital transformation in public service delivery.

User training programs will be developed to ensure that land administration staff can effectively operate the system. The system will include comprehensive documentation and help features to support self-learning. Change management strategies will address potential resistance to new processes and demonstrate the benefits of the digital system. Pilot implementation in selected offices will provide valuable lessons for broader deployment and enable iterative improvements based on actual operational experience.

1.8 Project Plan

1.8.1 Project Time Schedule

The project will be executed over a period of six months, divided into distinct phases with specific deliverables. The following table presents the detailed time schedule for the project activities:

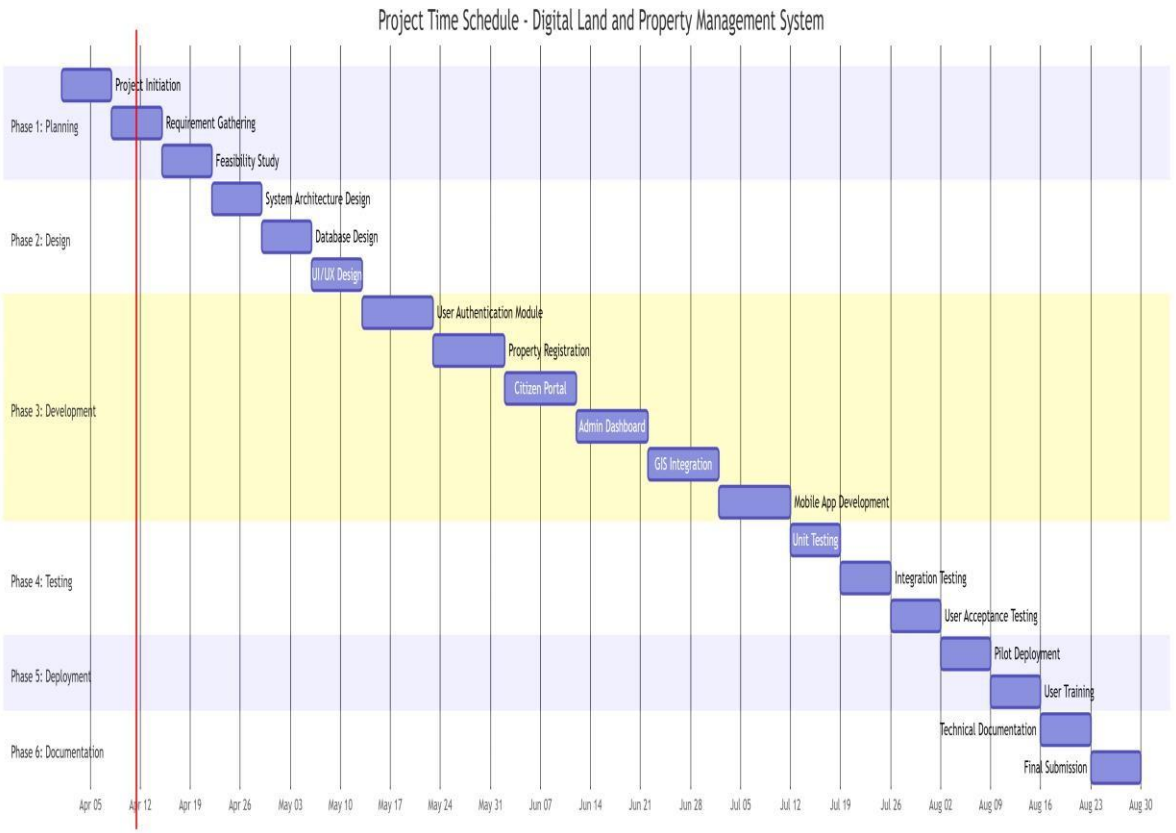


Figure 2 Gantt Chart for Project Time Schedule of the Digital Land and Property Management System

1.8.2 Budget Plan

The budget for this project has been estimated considering all necessary resources for successful completion. The following table presents the detailed budget breakdown:

Table 2 BUGET PLAN

No.	Item Description	Quantity	Estimated Cost (ETB)
1	Computer Hardware	2 units	156,000
2	Software Tools & Cloud Services	Various	65,000
3	Internet and Hosting	6 months	45,000
4	Domain Name & Deployment	Lump sum	85,000
5	Printing and Documentation	Lump sum	10,000
6	Transportation and Communication	6 months	12,000
7	Contingency (10%)	—	13,700
	TOTAL		**386,000 ETB**

CHAPTER TWO: REQUIREMENT ELICITATION

This chapter presents the findings of the requirements elicitation phase for the Digital Land and Property Management System (DLPMS). The primary objective of this chapter is to document the current state of land administration in Dire Dawa, to propose a digital solution that addresses identified deficiencies, and to specify the functional requirements of the proposed system using standard software engineering modeling techniques. The chapter begins with an overview of the existing system based on a site visit to the Dire Dawa Land Administration office, including both the legacy manual processes and the Cadastral Information System (CIS) that is currently being introduced. It then presents the proposed solution, justifies the preferred technical approach, and models the domain using CRC cards. Essential use case diagrams and documentation follow, and the chapter concludes with low-fidelity user interface prototypes that illustrate key system workflows. All requirements have been gathered through direct observation, stakeholder interviews, and document analysis, as described in Chapter One.

2.1 Overview of the Existing System

2.1.1 Site Investigation Method

To establish a factual basis for system requirements, the project team conducted an on-site investigation at the Dire Dawa Land Administration (DDLA) office. The investigation comprised direct observation of daily operations, inspection of physical record-keeping artifacts, and structured interviews with land officers, CIS system operators, and citizens awaiting services. The team also observed demonstrations of the existing Cadastral Information System (CIS) currently being deployed at the office. The following description is derived from primary evidence gathered during that visit, supplemented by a review of relevant procedural documents.

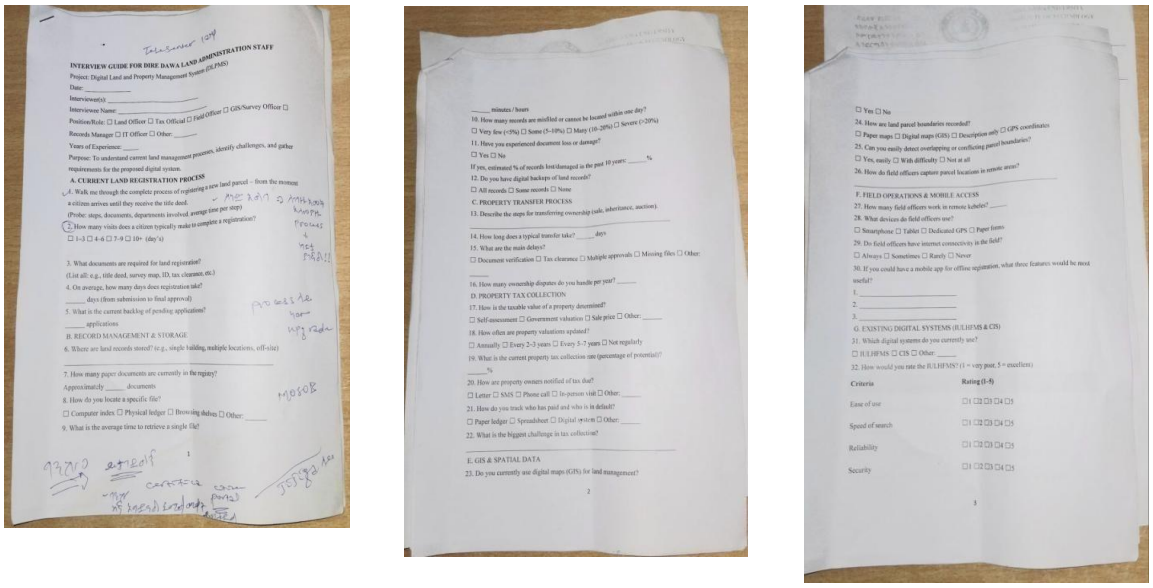


Figure 3Interview questions used during the site visit to the Dire Dawa Land Administration office

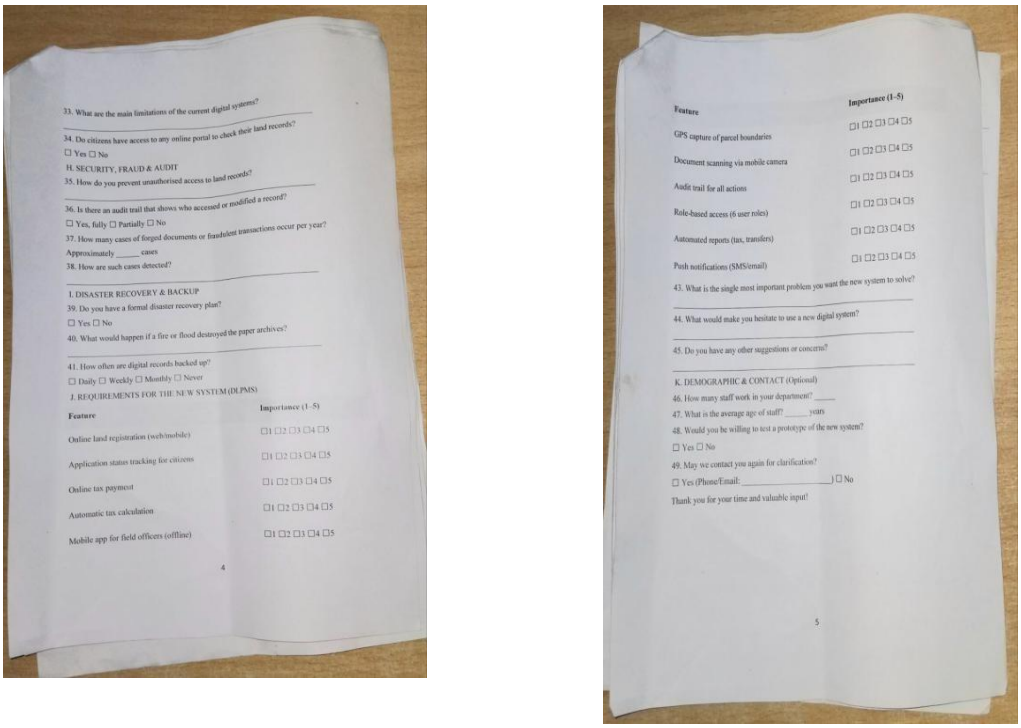


Figure 4 Additional interview sessions with land officers and citizens at DDLA

2.1.2 General Characteristics of the Current System

The land and property management system currently in use at the Dire Dawa Land Administration office is in a transitional phase between legacy manual operations and digital systems. Historically, all land-related records – including registration books, title deeds, transfer applications, tax ledgers, and dispute case files – were maintained exclusively as physical documents stored in metal filing cabinets and cardboard boxes within a dedicated archive room.

While this manual system still handles a significant portion of daily operations, the administration has recently begun deploying a Cadastral Information System (CIS) to modernize its workflows.

The CIS currently being introduced at DDLA is a powerful enterprise system designed to manage core land administration functions. It consists of four major operational modules: (1) the Archives module, which manages the digitization and storage of land documents; (2) the GIS module, which handles spatial data and land boundary mapping; (3) the BILE (Built-up Land Lease) module, which manages urban land lease administration; and (4) the Property Evaluation module, which assists in property valuation for tax and transfer purposes. This system represents a significant step toward digital transformation of land administration in Dire Dawa.

However, despite its capabilities, the CIS is still in the process of being fully adopted by the office staff. During the site visit, the team observed that many officers continue to rely on parallel manual processes alongside the CIS, and the system has several functional gaps that limit its effectiveness for citizen-facing services. Specifically, the CIS currently lacks a dedicated citizen portal for self-service access, does not support online payment integration, and does not provide mobile accessibility for citizens who need to interact with the land administration remotely. These limitations mean that citizens must still visit the office in person for most transactions, creating the same bottlenecks and delays that the digital transition was intended to eliminate.



Figure 5 Physical record storage and filing infrastructure observed at the DDLA archive room

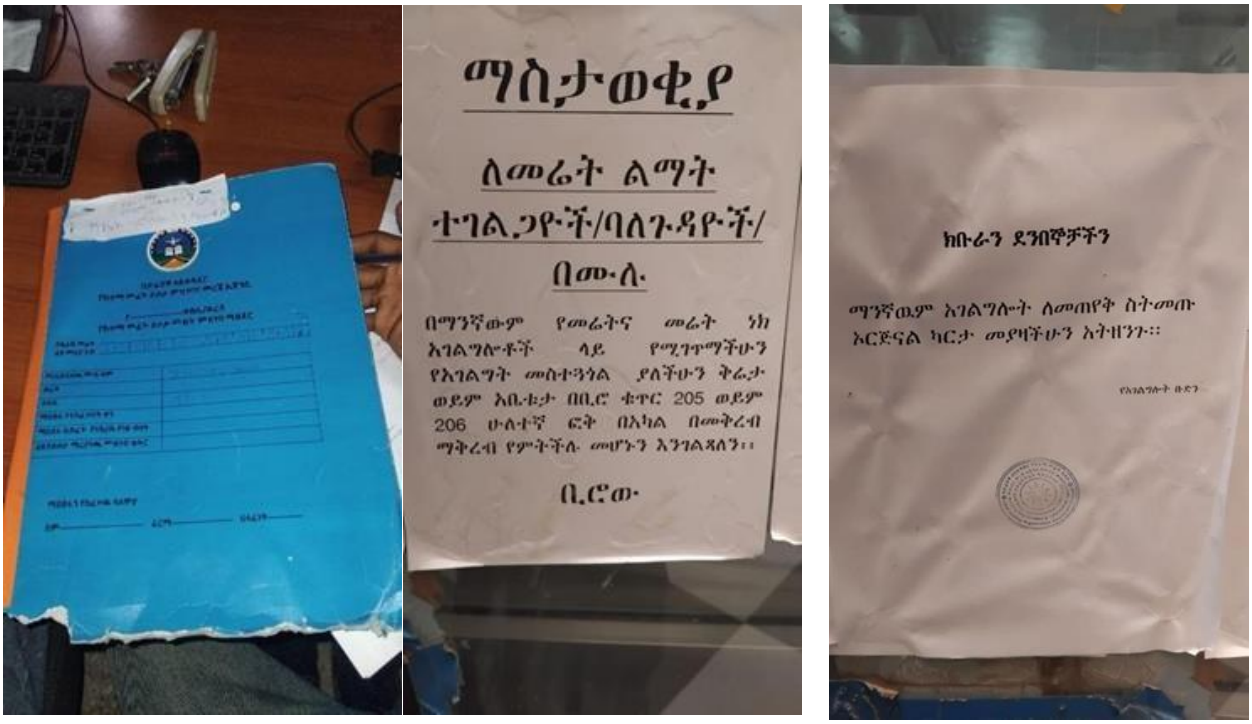


Figure 6 Additional views of the physical document storage and registration records at DDLA

2.1.3 Process Flows and Operational Deficiencies

While the CIS system has improved certain aspects of land administration at DDLA, several operational deficiencies persist due to the incomplete transition and functional gaps in the current digital infrastructure. The following observations were made during the site visit regarding both the legacy manual processes and the CIS operations:Manual registration processes persist alongside digital systems. Land registration entries are still performed by hand in bound registration books in parallel with CIS data entry, creating duplication of effort and increased risk of inconsistencies between the two record systems. Officers reported that this dual-entry process is time-consuming and that discrepancies between the physical and digital recor occasionally arise, requiring manual reconciliation.

figure 7 Land registration performed by hand in bound registration books alongside the CIS system And Complaint and service request process requiring physical visit to the bureau



Ownership transfers follow a partially automated procedure. A citizen submits a paper application together with the original title deed and a sale agreement. The land officer must enter the transfer details into the CIS system and simultaneously update the manual registration book. Due to the absence of automated cross-referencing between the two systems, duplicate transfer requests for the same parcel are not automatically detected. During the visit, officers reported that conflicting claims still arise, with some disputes remaining unresolved for extended periods.Record retrieval remains partially dependent on physical archives. Although the CIS Archives module enables digital lookup of recently digitized records, many older records exist only in physical form. An officer receiving a request may need to walk to the archive room, open filing cabinets, and visually scan file folders for records that have not yet been entered into the CIS. The team observed retrieval times for non-digitized records ranging from thirty minutes to over ninety minutes, and several files could not be located at all, with officers attributing the loss to misfiling or removal without proper documentation.

2.1.4 Specific Systemic Limitations

Despite the ongoing deployment of the CIS, the following specific limitations were identified through the site investigation. These limitations represent the gaps that the proposed DLPMS enhancement aims to address:

Incomplete digital coverage. The CIS Archives module has not yet digitized all historical

land records. A significant proportion of legacy documents, particularly those predating the CIS introduction, remain accessible only in physical form. This creates a two-tier system where recent records are digitally accessible while older records require manual retrieval, resulting in inconsistent service delivery and delays for citizens whose records fall in the non-digitized category.

No citizen-facing portal or self-service access. The current CIS is designed exclusively for internal use by land administration staff. Citizens cannot access the system to view their land records, check application statuses, or submit transfer requests online. Every interaction requires a physical visit to the DDLA office, which imposes significant time and cost burdens on citizens, particularly those residing in peri-urban areas or the surrounding rural districts.

No payment integration. The CIS does not support online or mobile payment processing for land-related fees, taxes, or service charges. All payments must be made in person at the office or through bank tellers, with paper receipts manually entered into the system. This creates additional visits and delays for citizens and increases the administrative workload for cash-handling officersLimited audit trail and accountability. While the CIS logs certain user actions, the logging is not comprehensive and does not capture all critical operations such as ownership transfers, record modifications, or boundary changes with sufficient detail. Old values and new values for critical data changes are not systematically recorded, making forensic investigation of irregularities difficult.

No mobile application for citizens. The CIS is accessible only through desktop workstations within the office network. There is no mobile application that would enable citizens to book appointments, make payments, submit documents, or track their application statuses remotely. This is a significant gap given the rapid growth of mobile phone penetration in Ethiopia and the increasing citizen expectation for mobile-first government services.

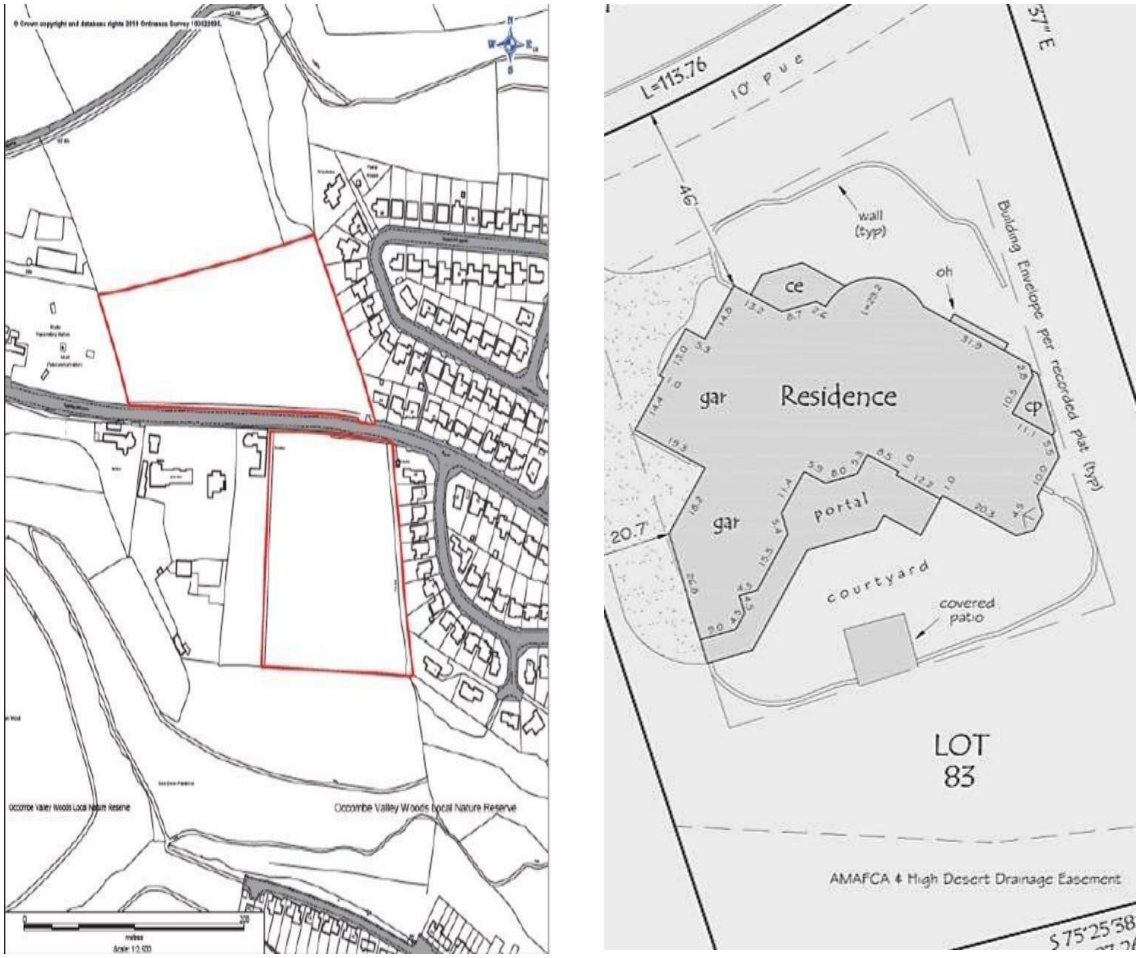


Figure 8 Hand-drawn property boundary sketches still used alongside the CIS GIS module

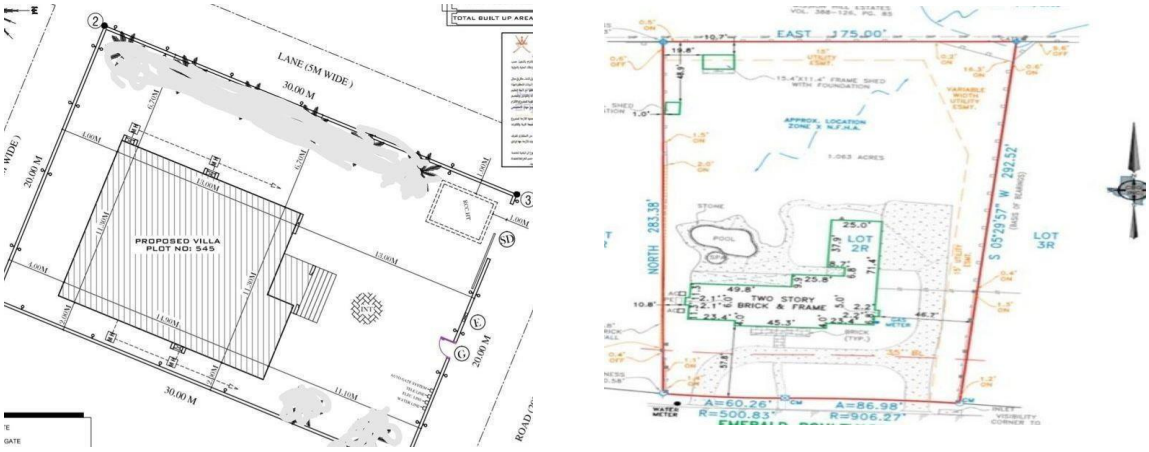


Figure 9 Additional views of the CIS system interface and land boundary documentation at DDLA

2.1.5 Conclusion of the Investigation

The land administration system at Dire Dawa is in a transitional state, with a powerful Cadastral Information System (CIS) being progressively deployed alongside legacy manual

processes. The CIS, with its four core modules covering Archives, GIS, BILE, and Property Evaluation, represents a strong foundation for digital land management. However, the investigation revealed significant gaps that limit the system's effectiveness, particularly the absence of a citizen-facing portal, lack of payment integration, no mobile application support, incomplete audit logging, and partial digitization of historical records. Citizens still experience long waiting times, repeated office visits, and manual paper-based procedures for many common transactions. The evidence gathered during the site visit confirms a clear need for system enhancements that build upon the existing CIS infrastructure to deliver a fully digital, citizen-centric land administration experience, as proposed in the subsequent sections of this chapter.

2.2 Proposed Solution

The proposed enhancement to the Digital Land and Property Management System will build upon the existing CIS infrastructure at DDLA by adding critical missing functionalities that will transform the system into a fully digital, citizen-centric platform. Rather than replacing the CIS, the proposed solution extends and complements it, leveraging the solid foundation already established by the four core CIS modules (Archives, GIS, BILE, and Property Evaluation) and filling the identified functional gaps.

The proposed system will introduce three major new capabilities. First, a Citizen Service Portal will be developed as both a web-based application and a mobile application, allowing landowners and citizens to access their land records, apply for ownership transfers, upload supporting documents, track application statuses, book appointments, and make online payments without visiting the DDLA office in person. Second, an integrated Payment Gateway will enable citizens to pay for land-related fees, property taxes, and service charges electronically through mobile money platforms such as telebirr and bank transfers, eliminating the need for physical payment visits. Third, a comprehensive Audit Logging system will be implemented to record every system action with full detail, including timestamps, user identities, IP addresses, and before-and-after values for all critical data modifications, ensuring complete transparency and accountability.

Land officers will benefit from streamlined workflows that reduce dual-entry requirements by providing unified interfaces for registration, transfer processing, and document verification. Tax officials will be able to manage property tax records with

automated calculations and integrated payment tracking. GIS officers will have enhanced boundary verification tools with automated overlap detection. System administrators will gain full visibility into all system operations through a comprehensive dashboard and audit log viewer.

The proposed system can be developed using different architectural approaches depending on the specific requirements and operational context. The following options were considered:

Web-based system: The system can be developed as a web-based application accessible via standard web browsers. This approach would enable land officers, tax officials, and administrators to access the system from any desktop computer with an internet connection, making it flexible and accessible for office-based operations. Citizens would access the portal through web browsers on their personal devices.

Mobile application: A dedicated mobile application for Android and iOS devices would provide citizens with a convenient and streamlined user experience for common transactions such as appointment booking, payment processing, application tracking, and document uploading. This approach would significantly reduce the need for physical office visits and extend service accessibility to citizens in peri-urban and rural areas around Dire Dawa.

Hybrid system: The system can be developed as a combination of both web-based and mobile applications, providing the flexibility and comprehensive functionality of a web portal for office use alongside the convenience and offline capabilities of a mobile application for citizens and field workers.

2.3 Preferred Solution

After evaluating the three possible development approaches described in Section 2.2, the hybrid system (web-based portal + mobile application) is selected as the preferred solution for the Digital Land and Property Management System. This choice is based on the specific operational needs of land administration in Dire Dawa, which require both robust office-based tools for land officers and accessible mobile interfaces for citizens.

The preferred hybrid solution will consist of two primary components:

A web-based administrative portal accessible via desktop browsers for land officers, tax officials, GIS officers, and system administrators. This portal will integrate with and extend

the existing CIS system, providing enhanced workflow management, comprehensive audit logging, and reporting capabilities.

A citizen-facing mobile application for Android devices that enables landowners to book appointments, view their land records, apply for ownership transfers, upload required documents, make electronic payments via integrated payment gateways (including telebirr), and track the status of their applications – all without needing to visit the DDLA office physically.

This solution is preferred because it provides the following advantages:

✓ Land officers can process registrations and transfers from their office computers through an enhanced web interface that integrates with the existing CIS, eliminating the need for dual manual-digital data entry.

✓ Citizens can perform most common land-related transactions from their mobile phones, including appointment booking, payment processing, and application tracking, dramatically reducing the number of required physical visits to the DDLA office.

✓ Online payment integration through mobile money platforms enables citizens to pay fees and taxes remotely, reducing queues at the office and improving revenue collection efficiency.

✓ Comprehensive audit logging across both the web portal and mobile application ensures full traceability of all system actions, supporting transparency and accountability objectives.

✓ The system remains accessible to citizens even in areas with intermittent connectivity, as the mobile application can cache essential information and synchronize when connectivity is restored.

✓ The hybrid approach leverages the existing CIS infrastructure while adding the missing citizen-facing and payment capabilities, representing a cost-effective enhancement rather than a complete system replacement.

2.4 Domain Modeling with CRC Cards

CRC (Class-Responsibility-Collaboration) cards are used to identify the main classes (entities) in the system, define what each class is responsible for, and specify which other

classes it collaborates with. The CRC cards below model the core domain entities of the Digital Land and Property Management System, including both the existing CIS components and the newly proposed modules.

Table 3 CRC Card 1: Land Parcel

Class	Responsibilities	Collaborators
Land Parcel	Store parcel ID, location, size, and land use type Store GIS boundary coordinates (polygon) Link to the current owner(s)	Owner, Land Registry, GIS Service
	Maintain historical ownership records Check boundary against neighboring parcels for overlaps	

Table 4 CRC Card 2: Owner

Class	Responsibilities	Collaborators
Owner	Store owner name, national ID, phone number, address Link to one or more land parcels owned Verify identity documents uploaded by citizen Maintain ownership history (date acquired, method)	Land Parcel, Citizen Portal, Authentication Service

Table 5 CRC Card 3: Land Registry

Class	Responsibilities	Collaborators

Land Registry	Register new land parcels into the system Process ownership transfer requests Issue digital land certificates (PDF with QR code) Search and retrieve records by parcel ID or owner name Generate reports (monthly registrations, transfers)	Land Parcel, Owner, Transfer Request, Audit Log

Table 6 CRC Card 4: Transfer Request

Class	Responsibilities	Collaborators
Transfer Request	Capture seller, buyer, parcel, sale price, and date Validate that the seller is the current owner Attach supporting documents (sale agreement, tax clearance) Track approval workflow (pending → tax check → officer review → approved/rejected)	Owner, Land Parcel, Tax Record, Workflow Engine

Table 7 CRC Card 5: Tax Record

Class	Responsibilities	Collaborators

Tax Record	Compute property tax based on parcel size, location, and land use Record tax payments with date and receipt number Maintain list of parcels with unpaid arrears Flag a parcel for tax hold (prevent transfer until paid)	Land Parcel, Owner, Payment Gateway

Table 8 CRC Card 6: User

Class	Responsibilities	Collaborators
User	Authenticate users using username/email and password (JWT token) Enforce role-based permissions (Citizen, Land Officer, Tax	Authentication Service, Audit Log, Role Permission Matrix
	Official, GIS Officer, Admin) Log all user actions (login, view, create, update, delete) Manage user profile (change password, update contact)	

Table 9 CRC Card 6: Audit Log

Class	Responsibilities	Collaborators

Audit Log	Record every create, update, delete, and view action Store timestamp, user ID, IP address, action type, affected record ID Store old value and new value for critical changes Provide tamper-evident log for security investigations	User, Land Parcel, Transfer Request, Land Registry

Table 10 CRC Card 7: GIS Service

Class	Responsibilities	Collaborators
GIS Service	Import parcel boundaries from shape files or GeoJSON files Render interactive map showing parcel polygons Automatically detect boundary overlaps or gaps Store and retrieve spatial data using PostGIS	Land Parcel, GIS Officer, Boundary Validator

Table 11 CRC Card 8: Payment Gateway

Class	Responsibilities	Collaborators

Payment Gateway	Process electronic payments via mobile money (telebirr) and bank transfer Generate digital payment receipts Record transaction details (amount, date, reference number) Handle payment failures and retries Reconcile payments with land service fees and tax records	Tax Record, Transfer Request, Citizen Portal, Land Registry

Table 12 CRC Card 10: Appointment Scheduler (New)

Class	Responsibilities	Collaborators
Appointment Scheduler	Manage available appointment slots for land office services Allow citizens to book, reschedule, or cancel appointments via mobile app Send appointment reminders via SMS notification Track appointment history and no-show statistics	Citizen Portal, User, Land Officer

These CRC cards represent the core domain model of the enhanced Digital Land and Property Management System. Cards 1 through 8 model entities that align with or extend the existing CIS system, while Cards 9 and 10 represent the new components being proposed: the Payment Gateway and Appointment Scheduler, which address the critical gaps identified during the site investigation.

2.5 Essential Use Case Diagram

A use case diagram shows the interactions between users (actors) and the system. It describes what the system does from the user's perspective, without showing how it is done internally.

The following actors interact with the enhanced Digital Land and Property Management System:

Citizen – A landowner or applicant who accesses the system through the mobile app or web portal to view records, apply for transfers, make payments, and book appointments.

Land Officer – A government employee who processes land registrations and transfer requests through the web-based administrative portal.

Tax Official – An employee who manages property tax records and payments, with access to integrated payment tracking.

GIS Officer – An employee who uploads and verifies land boundary maps using the GIS module.

System Administrator – A person who manages users, views audit logs, configures system settings, and performs system maintenance.

The essential use cases for each actor are listed below:

Citizen:

- ✓ Register account (mobile app / web portal)
- ✓ Login / Logout
- ✓ View my land records
- ✓ Apply for land transfer
- ✓ Upload supporting documents (sale agreement, ID copy)
- ✓ Track application status cancel or reschedule
- ✓ Pay fees and taxes via mobile money
- ✓ View payment history and download receipts

Land Officer:

- ✓ Process new land registration
- ✓ Verify ownership documents
- ✓ Approve transfer request

- ✓ Reject transfer request (with reason)
- ✓ Search land records (by parcel ID, owner name, location)
- ✓ Generate reports (daily, weekly, monthly)
- ✓ View appointment schedule

Tax Official:

- ✓ Calculate property tax for a parcel
- ✓ Record tax payment (date, amount, receipt)
- ✓ View list of parcels with tax arrears
- ✓ Flag parcel for tax hold (prevents transfer)
- ✓ Generate tax receipt (PDF)
- ✓ View integrated payment dashboard

GIS Officer:

- ✓ Upload parcel boundary (shape file, GeoJSON, or draw on map)
- ✓ Edit parcel geometry (adjust polygon)
- ✓ Verify boundary against neighboring parcels
- ✓ Report boundary overlap or gap

System Administrator:

✓ Manage user roles and permissions

✓ View full audit log (all system actions)

✓ Backup database

✓ Restore database from backup

✓ Configure system settings (tax rates, fees, appointment slots)

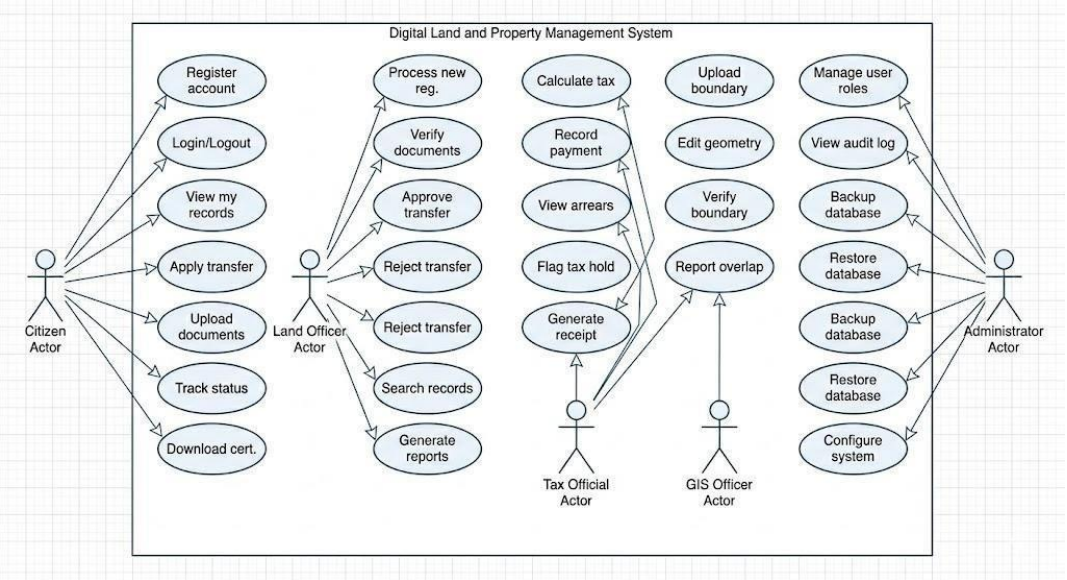


Figure 10: Essential Use Case Diagram of the Digital Land and Property Management System

2.6 Essential Use Case Documentation

This section provides detailed documentation for the six most important use cases of the enhanced Digital Land and Property Management System. Each use case includes the use case ID, name, actor, description, precondition, postcondition, basic flow, and alternative flows.

UC-01: Apply for Land Transfer

Table 13 UC-01: Apply for Land Transfer

Element	Description
Use Case ID	UC-01
Use Case Name	Apply for Land Transfer

Actor	Citizen (seller)
Description	The citizen initiates a request to transfer ownership of a land parcel to a buyer through the mobile application or web portal.
Precondition	The citizen is logged in and owns at least one land parcel registered in the system.
Postcondition	A transfer request is created with status "Pending Tax Check" and the citizen receives a confirmation notification.
Basic Flow	1. Citizen opens the mobile app or web portal and navigates to "Apply for Transfer." 2. System displays a list of parcels owned by the citizen. 3. Citizen selects the parcel to transfer. 4. System prompts for buyer details (name, national ID, phone). 5. Citizen enters buyer information. 6. Citizen uploads the sale agreement (PDF or image). 7. System validates that the seller is the current registered owner. 8. System creates a Transfer Request with a unique tracking ID. 9. System displays confirmation message with the tracking ID.
Alternative Flows	5a. Buyer national ID not found in the system → system allows manual entry. 6a. Missing or invalid document → system shows error message; citizen must upload the
	correct document. 7a. Seller is not the current owner → system rejects the request with an explanation.

Table 14 UC-02: Process Land Registration

UC-02: Process Land Registration

Element	Description
Use Case ID	UC-02
Use Case Name	Process Land Registration
Actor	Land Officer
Description	The land officer registers a new land parcel with owner details and boundary information through the administrative web portal.
Precondition	The applicant has submitted all required documents (physical or digital) and the officer is logged into the system.
Postcondition	New Land Parcel and Owner records are created in the system and a digital certificate is generated.
Basic Flow	1. Officer clicks "New Land Registration" on the dashboard. 2. System displays the registration form. 3. Officer enters parcel details (location, size, land use type). 4. Officer enters owner details (name, national ID, address, phone). 5. If a GIS boundary is available, the officer

	uploads or links it. 6. System checks for duplicate parcel ID or overlapping boundary. 7. Officer uploads the scanned deed and supporting documents. 8. System saves the parcel record and links it to the owner. 9. System generates a unique Parcel ID and a digital certificate (PDF). 10. Officer prints or emails the certificate to the applicant.
Alternative Flows	6a. Overlapping boundary detected → system displays an error and notifies the GIS officer for resolution. 8a. Missing mandatory field → system highlights the required field and prevents submission until completed.

Table 15 UC-03: Record Tax Payment

UC-03: Record Tax Payment

Element	Description
Use Case ID	UC-03
Use Case Name	Record Tax Payment
Actor	Tax Official
Description	The tax official records a property tax payment made by a landowner, either through the mobile payment gateway or in person.
Precondition	The landowner has paid the tax at the bank, via mobile money, or through the integrated payment gateway.

Postcondition	Tax record is updated with payment details. Tax hold (if applicable) is removed.
Basic Flow	1. Tax official searches for a parcel by ID or owner name. 2. System displays parcel details and current tax status. 3. Official clicks "Record Payment." 4. System prompts for amount, date, payment method, and receipt number. 5. Official enters the payment information. 6. System validates the payment amount against the outstanding due amount. 7. System updates the Tax Record. 8. If the parcel was on tax hold, the hold is automatically removed. 9. System generates a tax receipt (PDF).
Alternative Flows	6a. Partial payment → system records the partial amount and keeps the tax hold active until full payment is received. 6b. Overpayment → system records the actual amount and flags the excess as a credit balance.

UC-04: Upload Parcel Boundary

Table 16 UC-04: Upload Parcel Boundary

Element	**Description**
Use Case ID	UC-04
Use Case Name	Upload Parcel Boundary
Actor	GIS Officer
Description	The GIS officer uploads or draws the digital boundary of a land parcel using the GIS module.

Precondition	The parcel exists in the system (created by a Land Officer) and the officer is logged in.
Postcondition	Parcel geometry is stored in PostGIS; boundary status is set to "verified" if no overlap is detected.
Basic Flow	1. GIS officer searches for a parcel by ID. 2. System displays parcel information and the map interface. 3. Officer clicks "Upload Boundary." 4. Officer uploads a shape file, GeoJSON file, or draws a polygon directly on the interactive map. 5. System automatically checks for overlap with neighboring parcels. 6. If no overlap is detected, the system saves the geometry. 7. System sets the boundary status to "verified." 8. Officer confirms and exits.
Alternative Flows	5a. Overlap detected → system highlights the conflicting boundary and alerts the officer; the issue must be resolved with the Land Officer before proceeding.

UC-05: View Audit Logs

Table 17 UC-05: View Audit Logs

Element	Description
Use Case ID	UC-05
Use Case Name	View Audit Logs
Actor	System Administrator

Description	The administrator views a chronological record of all system actions across both the web portal and mobile application.
Precondition	Administrator is logged in with "Admin" role.
Postcondition	Filtered audit logs are displayed on screen or exported.
Basic Flow	1. Admin selects "Audit Logs" from the admin panel. 2. System displays the most recent log entries. 3. Admin applies filters (date range, user, action type, record ID). 4. System refreshes the list based on the applied filters. 5. Admin clicks "Export CSV" to download the filtered logs.
Alternative Flows	3a. No logs match the filter criteria → system displays "No records found" message.

UC-06: Book Appointment and Pay via Mobile App

Table 18 UC-06: Book Appointment and Pay via Mobile App

Element	Description
Use Case ID	UC-06
Use Case Name	Book Appointment and Pay via Mobile App
Actor	Citizen
Description	The citizen books an appointment with a land officer and pays the required service fee through the mobile application using mobile money.
Precondition	The citizen is registered and logged into the mobile app.

Postcondition	An appointment is scheduled, the service fee is paid, and the citizen receives a digital receipt and appointment confirmation via SMS.
Basic Flow	1. Citizen opens the mobile app and selects "Book Appointment." 2. System displays available dates and time slots for the selected service type. 3. Citizen selects a preferred date and time slot. 4. System displays the service fee amount and available payment methods. 5. Citizen selects mobile money (telebirr) and confirms the payment. 6. System redirects to the payment gateway for authentication. 7. Upon successful payment, the system confirms the appointment. 8. Citizen receives an SMS notification with appointment details and a digital receipt. 9. The appointment appears on the land officer's
	schedule dashboard.
Alternative Flows	7a. Payment fails → system informs the citizen and allows retry; the appointment slot remains reserved for a limited time. 2a. No available slots → system displays the next available date and offers a waitlist option.

2..7 Essential User Interface Prototype

This section presents low-fidelity prototypes of the main screens for the enhanced Digital Land and Property Management System. The prototypes illustrate both the administrative web portal interfaces used by officers and the mobile application interfaces used by citizens. These layouts show the placement of key elements such as headers, navigation menus, buttons, tables, maps, and payment interfaces. In the final system, these will be replaced with high-fidelity designs implemented using React.js and Tailwind CSS.

The following screens are included:

Screen 1: Citizen Mobile Dashboard – The main screen of the mobile application showing the citizen's land parcels, quick action buttons for appointment booking, payment, and application tracking.

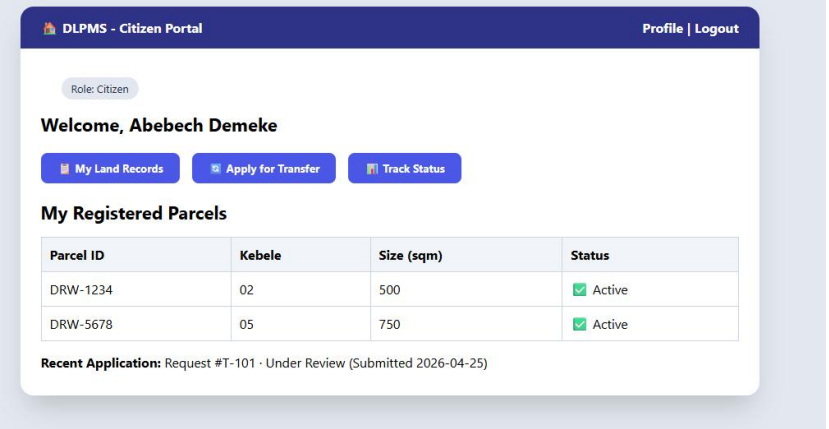


Figure 11: Citizen Mobile Dashboard – main screen showing land parcels and quick action buttons

Screen 2: Land Officer – Pending Transfers – The administrative web portal screen where land officers review and process pending transfer requests from citizens.

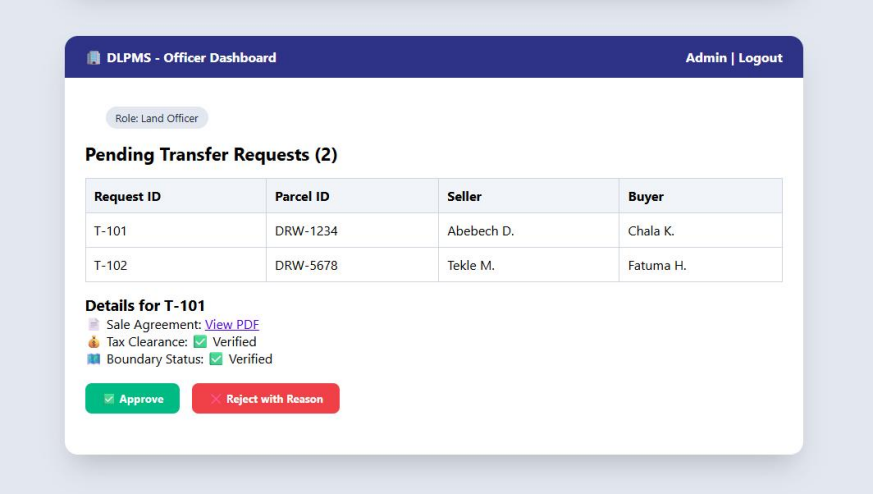


Figure 12 Land Officer – Pending Transfer Requests screen on the administrative web portal

Screen 3: GIS Officer – Boundary Verification – The GIS module screen where officers upload, verify, and edit land parcel boundaries on an interactive map.

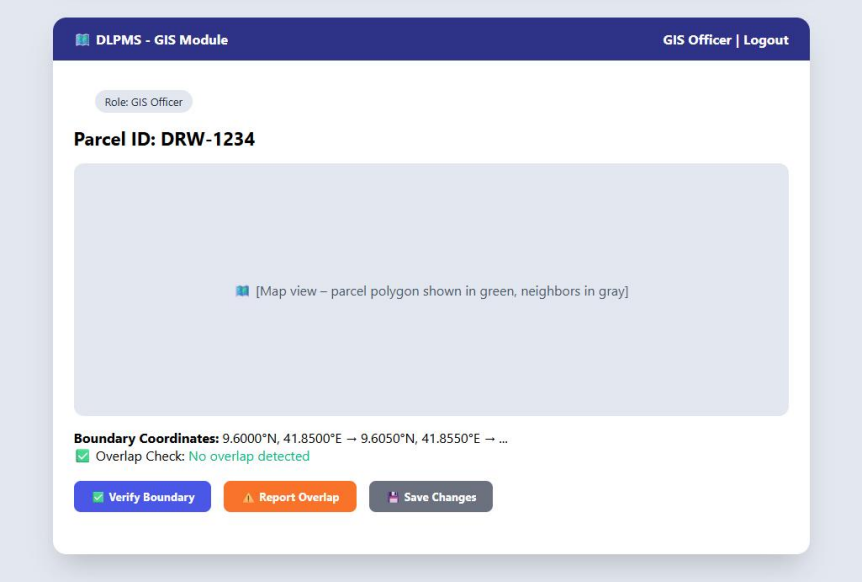


Figure13: GIS Officer – Boundary Verification screen with interactive parcel map

Screen 4: System Administrator – Audit Log Viewer – The administration screen for viewing all system actions with filtering and export capabilities.

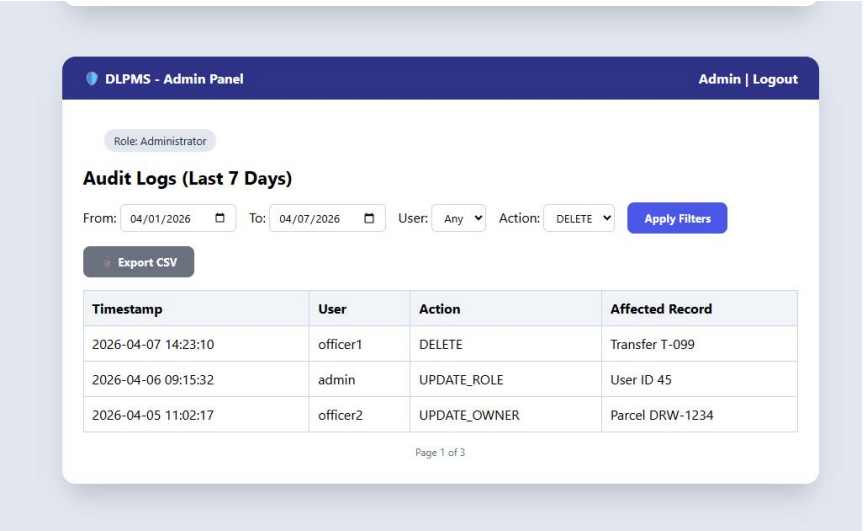


Figure 14 System Administrator – Audit Log Viewer with filtering and CSV export

CHAPTER THREE: SYSTEM ANALAYSIS

This chapter presents a comprehensive system analysis for the Digital Land and Property Management System (DLPMS). It begins with an overview of the proposed system and its relationship with the existing Cadastral Information System (CIS) at the Dire Dawa Land Administration (DDLA). The chapter then details both the functional and non-functional requirements elicited through stakeholder interviews, document analysis, and observation studies as described in Chapter Two. Finally, system modeling using UML diagrams—including use case diagrams, sequence diagrams, activity diagrams, and a class diagram—is presented to provide a visual specification of the system's behavior and structure.

3.1 Overview of the Proposed System

The Digital Land and Property Management System (DLPMS) is designed to enhance and extend the capabilities of the existing Cadastral Information System (CIS) currently operational at the Dire Dawa Land Administration (DDLA). The existing CIS consists of four primary modules: the Archives module for document management, the GIS module for spatial data visualization, the BILE (Building Inspection and Land Evaluation) module for property assessment, and the Property Evaluation module for valuation purposes. While the CIS has served as the backbone of land administration operations, it lacks citizen-facing digital services, automated payment integration, and comprehensive audit capabilities.

The DLPMS introduces four new components that integrate seamlessly with the existing CIS infrastructure:

1. Citizen Service Portal: A responsive web application and mobile app built with React.js and Tailwind CSS that allows citizens to register, view their land records, apply for land transfers, upload supporting documents, track application status, download certificates, and search public land records. The portal is designed as a Progressive Web App (PWA) to ensure offline-capable access on mobile devices.

2. Integrated Payment Gateway: A telebirr mobile money integration module that enables citizens to make property tax payments directly through the DLPMS. The payment gateway handles secure transaction processing, webhook signature validation, automatic receipt generation, and real-time status updates. This eliminates the need for citizens to physically visit the tax office for routine payments.

3. Comprehensive Audit Logging: A tamper-evident logging system that records every CRUD operation performed within the DLPMS. Each audit entry includes a cryptographic hash chain to ensure immutability, along with metadata such as the timestamp, user ID, IP address, action type, and before/after snapshots of modified data. This component is critical for regulatory compliance and accountability.

4. Appointment Scheduler: An online booking system that allows citizens to schedule appointments with land officers for in-person document verification and consultations. The scheduler supports time-slot management, automated SMS and email reminders sent 24 hours before the scheduled appointment, and integration with the officer's calendar to prevent double-booking.

The technology stack selected for the DLPMS comprises React.js with Tailwind CSS for the frontend user interface, Node.js with Express.js for the backend REST API server, PostgreSQL with the PostGIS spatial extension for the database layer, and Leaflet.js for interactive map rendering. Authentication is implemented using JSON Web Tokens (JWT) with role-based access control (RBAC) to ensure that users can only access features appropriate to their assigned roles.

The new system integrates with the existing CIS database through a RESTful API layer, enabling seamless data exchange between the legacy system and the modern web-based portal. This integration approach ensures that the DLPMS can leverage the existing spatial data, land records, and property information stored in the CIS without requiring a complete system migration, thereby reducing implementation risk and cost while maintaining operational continuity.

3.2 System Requirements

This section presents the complete set of system requirements for the DLPMS, categorized into functional requirements (what the system must do) and non-functional requirements (how the system should behave). These requirements were elicited from stakeholders including land officers, tax officials, GIS officers, system administrators, and citizens through the methods described in Chapter Two.

3.2.1 Functional Requirements

A total of 36 functional requirements have been identified for the DLPMS. These requirements

are organized by the primary actor responsible for initiating or interacting with each function. Table 3.1 presents the complete functional requirements specification with their associated actors and priority levels.

Table 19 : Functional Requirements Specification

Req ID	Requirement Description	Actor	Priority
FR-01	Register as a citizen with personal details and phone number	Citizen	High
FR-02	Login using phone number or email with password and OTP verification	Citizen	High
FR-03	View land records and property details associated with their account	Citizen	High
FR-04	Apply for land ownership transfer by submitting required documents	Citizen	High
FR-05	Upload supporting documents (ID, photos, forms) in PDF or image format	Citizen	High
FR-06	Track the status of submitted applications in real-time	Citizen	Medium

FR-07	Download digitally signed land ownership certificates in PDF format	Citizen	High
FR-08	Book appointments with land officers for in-person verification	Citizen	Medium
FR-09	Make property tax payments via telebirr mobile money integration	Citizen	High
FR-10	Search public land records by parcel ID, owner name, or location	Citizen	Medium
FR-11	Process new land registration applications submitted by citizens	Land Officer	High
FR-12	Verify uploaded documents for authenticity and completeness	Land Officer	High
FR-13	Approve land transfer requests after verification and validation	Land Officer	High
FR-14	Reject land transfer requests with documented reasons	Land Officer	High

FR-15	Generate comprehensive reports on land registrations and transfers	Land Officer	Medium
FR-16	Search and retrieve land records from the CIS database	Land Officer	Medium
FR-17	Calculate property tax based on land area, location, and property type	Tax Official	High
FR-18	Record manual offline payments received at the office counter	Tax Official	High
FR-19	View a dashboard with tax collection summaries and outstanding dues	Tax Official	Medium
FR-20	Generate official tax receipts for both online and offline payments	Tax Official	High
FR-21	Generate tax compliance reports by period and sub-city	Tax Official	Medium
FR-22	Upload and update land parcel boundary data from survey measurements	GIS Officer	High
FR-23	Verify parcel boundary accuracy against GIS cadastral maps	GIS Officer	High

FR-24	View and interact with parcel maps using the web-based GIS interface	GIS Officer	Medium
FR-25	Manage user accounts including creation, role assignment, and deactivation	System Admin	High
FR-26	View and filter comprehensive audit logs of all system activities	System Admin	Medium
FR-27	Configure system parameters including tax rates, fees, and notification templates	System Admin	Medium
FR-28	Perform database backup and restore operations	System Admin	High
FR-29	Authenticate all users via JWT tokens with role-based access control	System	High
FR-30	Send SMS notifications for appointment reminders and status updates	System	Medium

FR-31	Send email notifications for application status changes	System	Medium
FR-32	Generate PDF certificates with digital signatures and watermarks	System	High
FR-33	Validate webhook signatures for secure payment gateway callbacks	System	High
FR-34	Log all CRUD operations with timestamp, user, and action details	System	High
FR-35	Handle concurrent edits with optimistic locking and conflict detection	System	Medium
FR-36	Schedule automated appointment reminders 24 hours before the scheduled time	System	Medium

The functional requirements are prioritized based on their criticality to the core business processes of land administration. High-priority requirements (FR-01 through FR-05, FR-11 through FR-14, FR-17, FR-18, FR-22, FR-25, FR-28, FR-29, FR-32 through FR-34) represent features that are essential for the system to fulfill its primary purpose of enabling digital land and property management. Medium-priority requirements represent features that enhance usability and operational efficiency but are not critical for the minimum viable product.

3.2.2 Non-Functional Requirements

Thirteen non-functional requirements have been identified to ensure that the DLPMS meets the necessary quality attributes for a government land administration system. These requirements address performance, security, scalability, availability, and other cross-cutting concerns. Table

3.2 presents the complete non-functional requirements specification with measurable acceptance criteria.

Table 20 : Non-Functional Requirements Specification

Req ID	Category	Description	Acceptance Criteria
NFR-01	Performance	System response time shall be under 3 seconds for 95% of all requests under normal load conditions.	Response time measured via APM tools; 95th percentile < 3s for 500 concurrent users.
NFR-02	Security	All communications must use HTTPS encryption with TLS 1.2+ enforced; penetration test passes JWT authentication and bcrypt password hashing (cost factor 12).	OWASP Top 10 compliance; passwords stored as bcrypt hashes.
NFR-03	Scalability	The system shall support 500+ concurrent users without performance degradation.	Load test with 500+ simulated concurrent users shows < 10% response time increase.
NFR-04	Availability	The system shall maintain 99.5% uptime, allowing a maximum of 43.8 hours downtime per year.	Monitoring via uptime checker; monthly availability reports confirm >= 99.5%.
NFR-05	Usability	The citizen portal shall have responsive design compatible with Chrome, Firefox, and Safari (latest 2 versions).	Manual testing on specified browsers; layout adapts correctly on desktop, tablet, and mobile.

NFR-06	Data Integrity	All database operations must be ACID-compliant using PostgreSQL transactions.	Transaction rollback verified for failed operations; no partial updates observed.
NFR-07	Audit Trail	All CRUD operations must be logged with tamper-evident audit entries including timestamp, user ID, and action type.	Audit logs are immutable; hash chain verification passes; 100% of operations logged.
NFR-08	Backup	Automated daily database backups must be maintained with a 30-day retention policy.	Backup cron job verified; restore test passes monthly; retention enforced by automated cleanup.
NFR-09	Mobile Compat.	The citizen portal shall be deployable as a Progressive Web App (PWA) for offline-capable mobile access.	Lighthouse PWA audit score >= 90; service worker installs correctly on mobile devices.
NFR-10	API Docs	All REST API endpoints must be documented using OpenAPI/Swagger 3.0 specification.	Swagger UI accessible at /api-docs; all endpoints documented with request/response schemas.
NFR-11	Localization	The system shall support Amharic and English	All UI labels, messages, and system notifications

		languages with user-selectable language preference.	rendered correctly in both languages.
NFR-12	Accessibility	The citizen portal shall comply with WCAG 2.1 Level AA accessibility standards.	Automated Axe audit passes with 0 violations; manual screen-reader testing completed.
NFR-13	Data Export	All reports and audit logs must be exportable in CSV and PDF formats.	Export buttons generate correctly formatted CSV and PDF files matching displayed data.

The non-functional requirements are designed to ensure that the DLPMS meets the operational standards expected of a government digital service. The security requirements (NFR-02) align with the OWASP Top 10 security guidelines, while the availability target of 99.5% (NFR-04) accounts for scheduled maintenance windows. The localization requirement (NFR-11) is particularly important for the Dire Dawa context, where both Amharic and English are widely used in official communications.

3.3 System Modeling

This section presents the UML-based system models for the DLPMS. Use case diagrams provide a high-level overview of system functionality from the actors' perspectives, while sequence diagrams illustrate the message flows for critical use cases. Activity diagrams model the step-by-step workflows for key business processes, and the class diagram defines the static structure of the system's object-oriented design. These models were developed based on the CRC cards and use case descriptions documented in Chapter Two.

3.3.1 Use Case Diagram

The use case diagram for the DLPMS, shown in Figure 3.1, provides a comprehensive visual representation of all 28 use cases and the 5 primary actors that interact with the system. The five actors are: Citizen, Land Officer, Tax Official, GIS Officer, and System Admin. The Citizen actor has the highest number of associated use cases (10), reflecting the system's emphasis on providing accessible digital services to the public. Each use case is represented as an ellipse, and actors are depicted as stick figures positioned outside the system boundary.

The diagram includes five include and extend relationships between use cases. The include relationships represent mandatory behavioral inclusions—for example, the "Authenticate User" use case is included in "Apply for Transfer" because authentication is a prerequisite for any transaction. The extend relationships represent optional or conditional behaviors—for example, "Generate Tax Receipt" extends "Make Payment" and is only triggered when a payment is successfully completed. Table 3.3 summarizes these relationships with their descriptions.

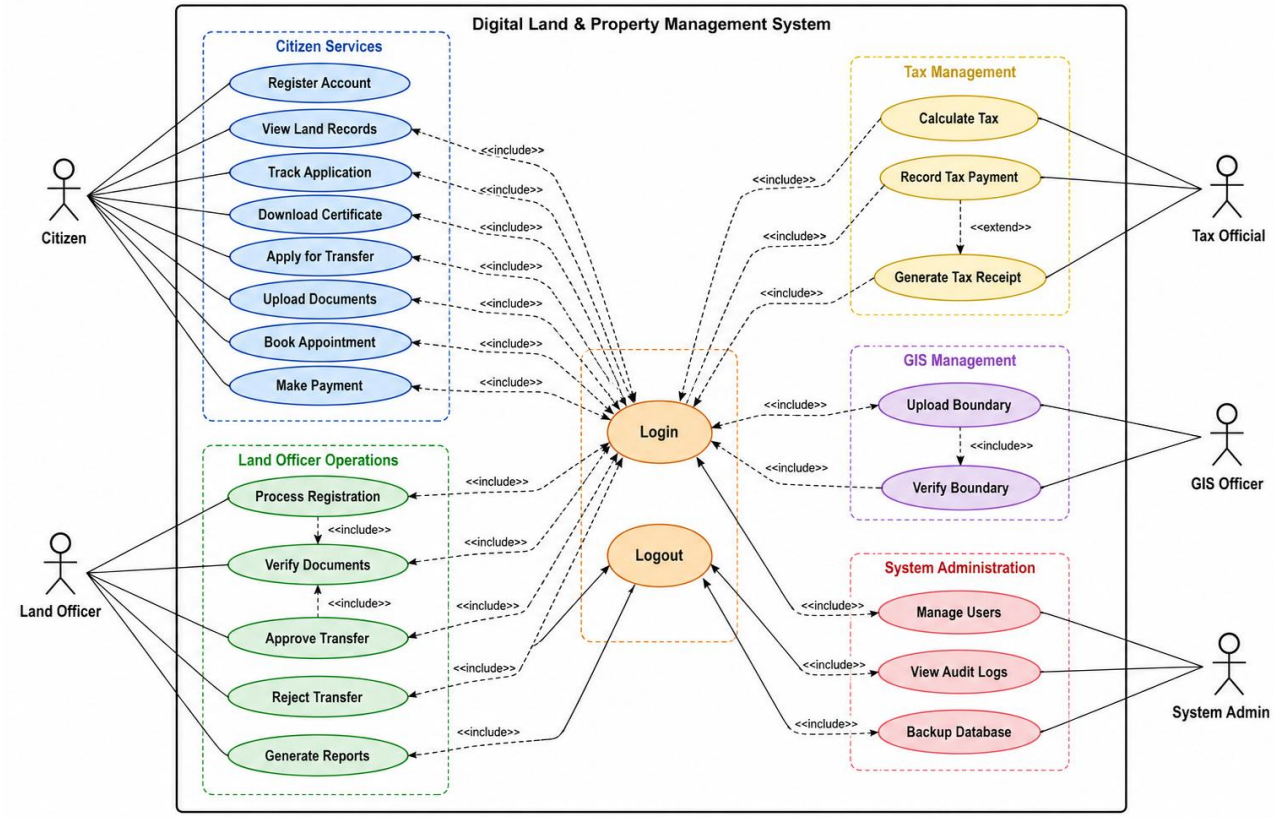


Figure 15:System Use Case Diagram

Table 3.3: Include and Extend Relationships in Use Case Diagram

Table 21 Include and Extend Relationships in Use Case Diagram

Ref ID	Base Use Case	Relationship	Included/ Extended Use Case	Description

UC-01	Apply for Transfer	<<include>>	Authenticate User	Every transfer application requires the citizen to be authenticated first.
UC-02	Make Payment	<<extend>>	Generate Tax Receipt	A tax receipt is automatically generated only when a payment is completed successfully.
UC-03	Approve Transfer	<<include>>	Verify Documents	Document verification is a mandatory step within the transfer approval workflow.
UC-04	Book Appointment	<<include>>	Send SMS Notification	An SMS confirmation is sent to the citizen whenever an appointment is booked.
UC-05	Process Registration	<<extend>>	Upload Boundary	Boundary data upload by the GIS Officer may be triggered during the registration process.

3.3.2 Sequence Diagrams

Sequence diagrams illustrate the dynamic behavior of the DLPMS by showing the chronological sequence of messages exchanged between objects for specific use case scenarios. The following two sequence diagrams model the two most critical workflows identified during requirements elicitation: the citizen payment process and the land transfer approval process.

3.3.2.1 Citizen Makes Payment via telebirr

Figure 3.2 presents the sequence diagram for the “Citizen Makes Payment” use case. This diagram models the complete message flow between five participating objects: the Citizen actor, the DLPMS Backend, the telebirr Payment Gateway, the Tax Record Database, and the Notification Service.

The interaction begins when the Citizen initiates a payment request by selecting a parcel and confirming the tax amount (Step 1). The DLPMS Backend validates the request and generates a unique transaction reference before forwarding it to the telebirr Payment Gateway (Step 2). The Payment Gateway returns a payment URL and USSD push prompt to the citizen’s mobile device (Step 3). The Citizen completes the payment on their phone (Step 4), and the Payment Gateway sends an asynchronous webhook callback to the DLPMS Backend with the payment result (Step 5). The Backend validates the webhook signature to prevent spoofing (Step 6) and, upon successful validation, updates the Tax Record Database (Step 7). The Notification Service is then invoked to send an SMS confirmation to the citizen (Step 8) and an email receipt (Step 9). Finally, the DLPMS Backend returns a success response to the Citizen’s browser session (Step 10).

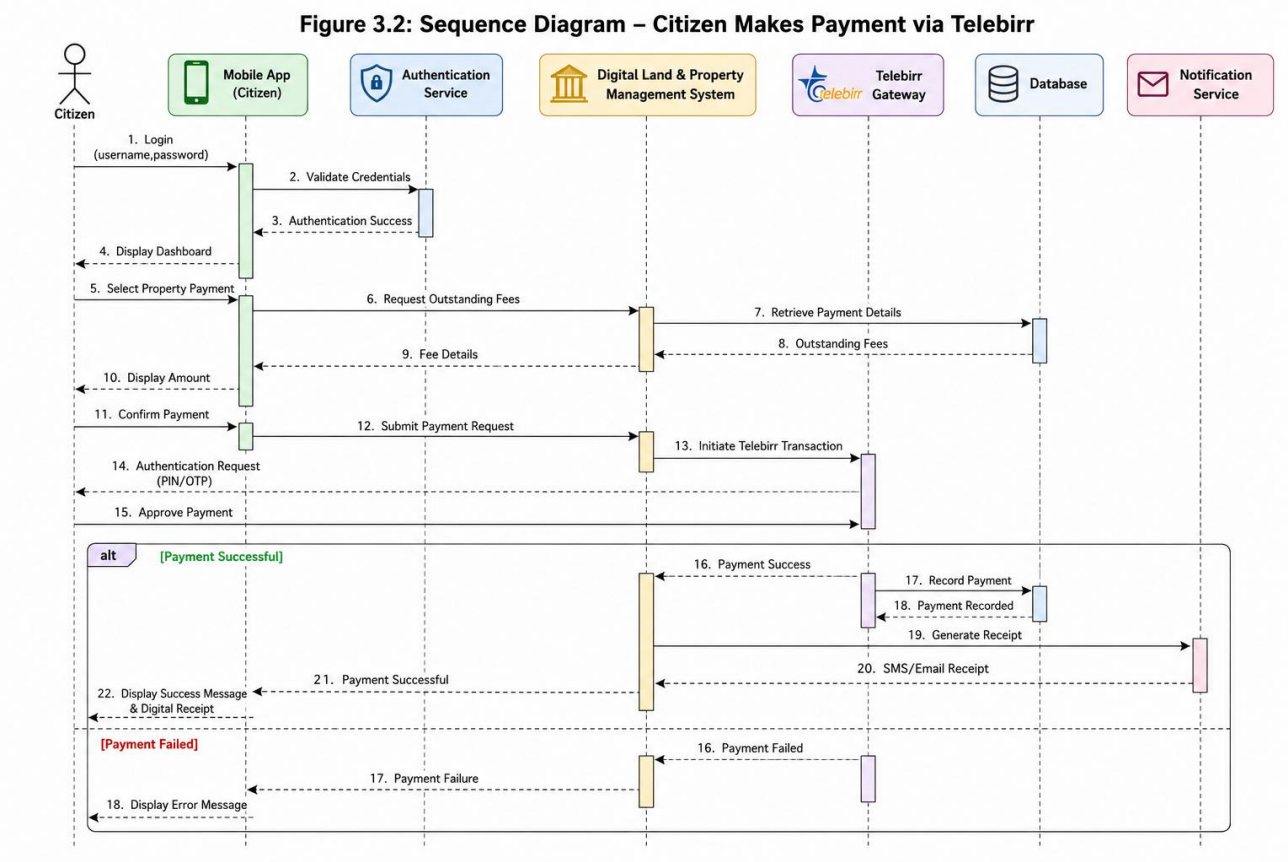


Figure 16 Sequence Diagram – Citizen Makes Payment via telebirr

3.3.2. 2 Land Officer Approves Transfer

Figure 3.3 depicts the sequence diagram for the "Land Officer Approves Transfer" use case. This diagram models the message flow between five objects: the Land Officer actor, the DLPMS Backend, the CIS Database, the Audit Log, and the Notification Service.

The workflow begins when the Land Officer reviews a pending transfer request from the application queue (Step 1) and selects the request for processing (Step 2). The DLPMS Backend retrieves the associated documents and parcel details from the CIS Database (Step 3). The Land Officer verifies the uploaded documents against the transfer requirements (Step 4) and submits an approval decision with comments (Step 5). The Backend performs business rule validation, including tax clearance verification and buyer eligibility checks

(Step 6). Upon successful validation, the Backend updates the land ownership record in the CIS Database (Step 7) and creates an immutable audit log entry recording the approval action with the officer's ID, timestamp, and before/after ownership state (Step 8). The Notification Service is triggered to send SMS and email notifications to both the seller and buyer (Step 9). Finally, a success confirmation is returned to the Land Officer's dashboard (Step 10).

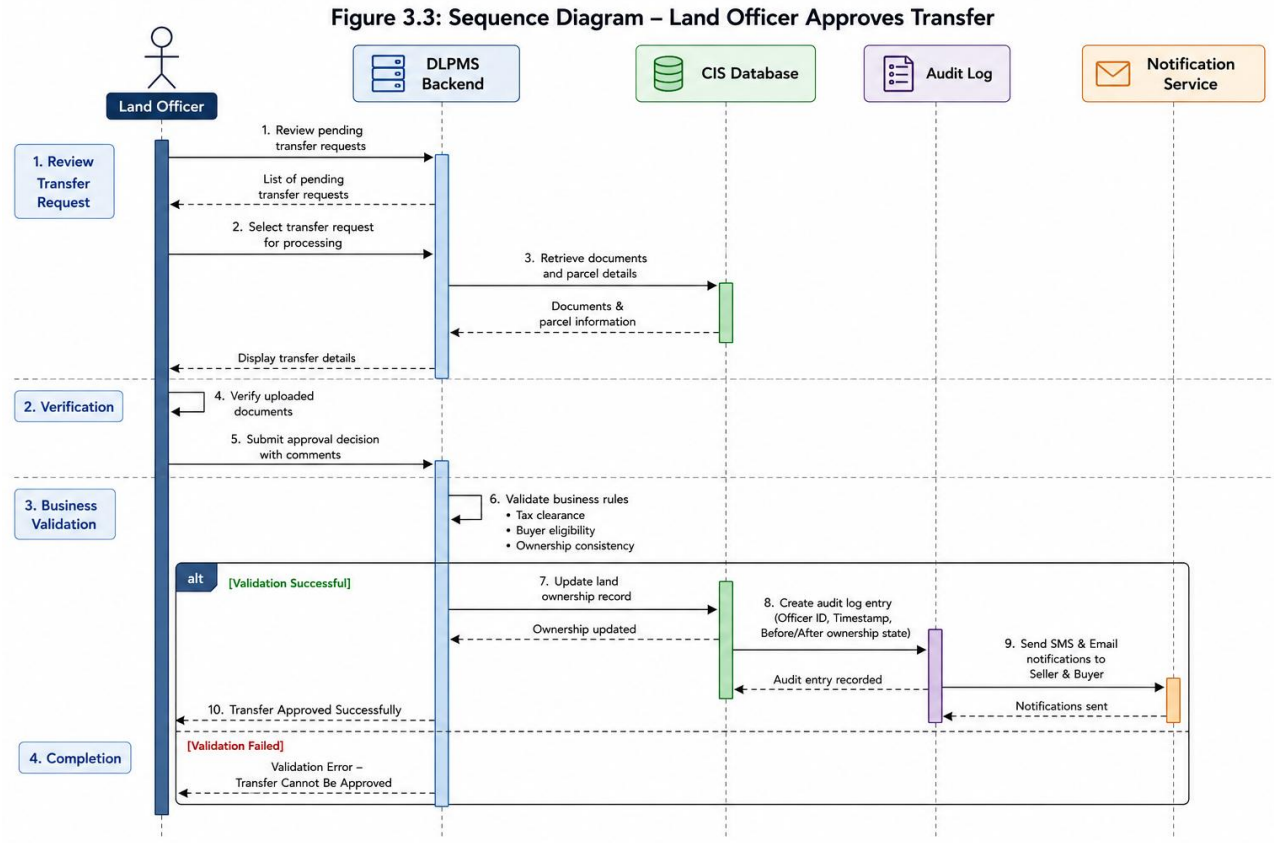


Figure 17 : Sequence Diagram – Land Officer Approves Transfer

3.3.3 Activity Diagrams

Activity diagrams model the step-by-step workflows of key business processes in the DLPMS, including decision points, parallel activities, and loop-back mechanisms. The following two activity diagrams illustrate the land transfer application process from the citizen's perspective and the appointment booking workflow.

3.3.3.1 Land Transfer Application Process

Figure 3.4 presents the activity diagram for the Land Transfer Application Process. This diagram models the complete citizen-facing workflow from initial authentication through to application

submission and status tracking.

The workflow includes two critical decision points. The first decision node, labeled “Tax Hold Check,” determines whether the parcel has any outstanding tax obligations. If a tax hold exists, the citizen is redirected to the payment page and must clear the outstanding balance before proceeding with the transfer application. The second decision node, labeled “Buyer Validation,” verifies that the prospective buyer’s identity documents match the submitted information. If validation fails, the citizen is presented with an error message and given the option to correct and resubmit the buyer details, creating a loop-back mechanism. An early termination path is also modeled for cases where the citizen voluntarily abandons the application before submission.

The fork and join nodes in the diagram represent parallel activities that occur during document upload, where the citizen can simultaneously upload multiple supporting documents (identification, photographs, and signed forms) before the system performs bulk validation and proceeds to the final submission step.

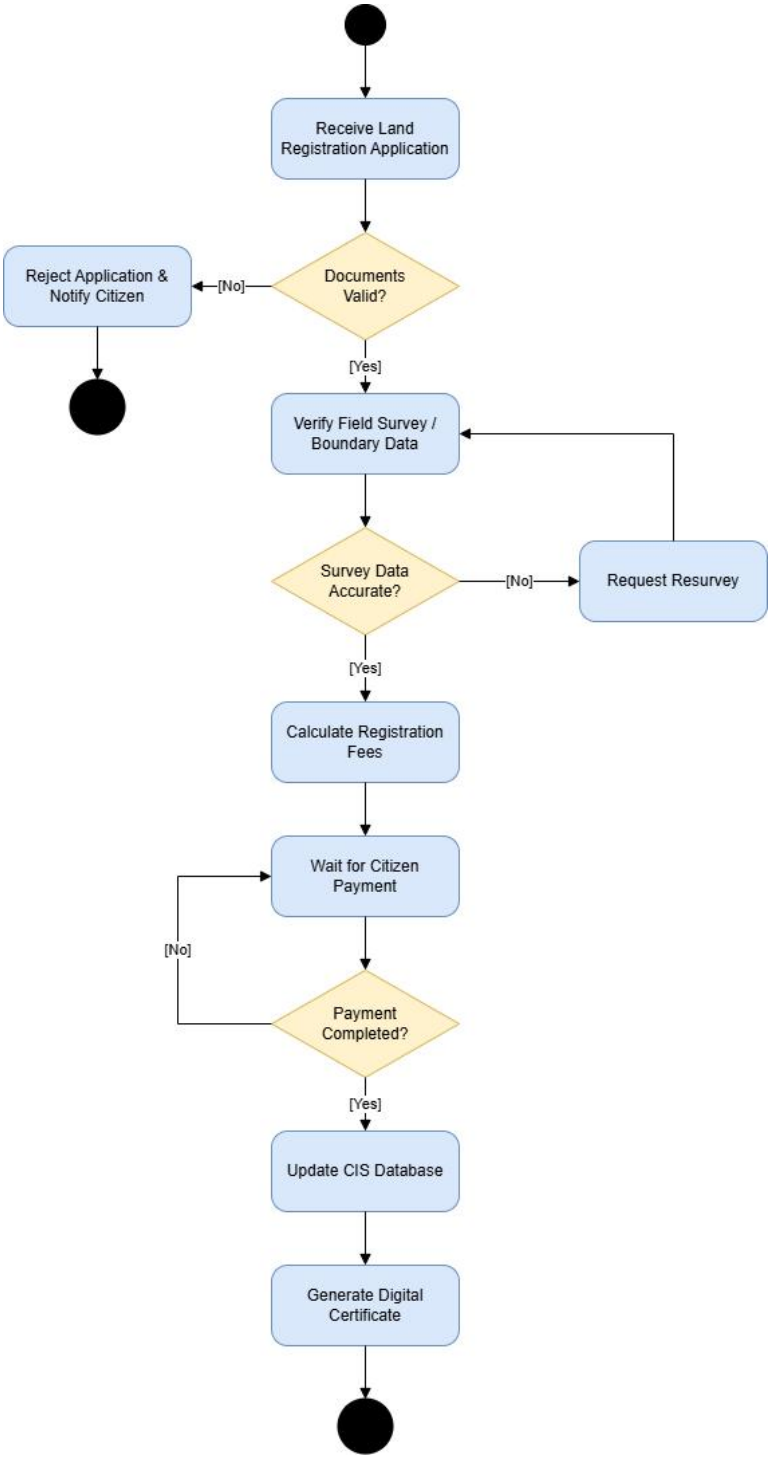


Figure 18 :Activity Diagram – Land Transfer Application Process

3.3.3.2 Book Appointment Process

Figure 3.5 illustrates the activity diagram for the Book Appointment workflow. This diagram models the complete citizen-facing process for scheduling an in-person appointment with a land officer through the DLPMS.

The workflow begins when a citizen navigates to the appointment booking section after authenticating into the system. The first action involves selecting the type of service required—such as document verification, land transfer consultation, or property assessment review. Once the service type is selected, the system queries the available time slots for the selected officer and date range. A decision node then checks whether any slots are available. If no slots are available, the citizen is presented with alternative dates or the option to join a waitlist for cancellations.

Upon selecting a preferred time slot, the system performs a real-time availability check to prevent double-booking. If the slot is still available, the citizen confirms the booking by providing the purpose of the visit and any supplementary details. The system then creates the appointment record and triggers an automated notification chain: an SMS confirmation is sent to the citizen's registered phone number, and an email confirmation with appointment details is dispatched simultaneously. A second notification is scheduled for 24 hours before the appointment as a reminder.

The diagram also models an alternative path where the citizen wishes to reschedule or cancel an existing appointment. In this case, the system verifies whether the cancellation falls within the permissible time window and, upon successful cancellation, releases the time slot back to the available pool for other citizens to book.

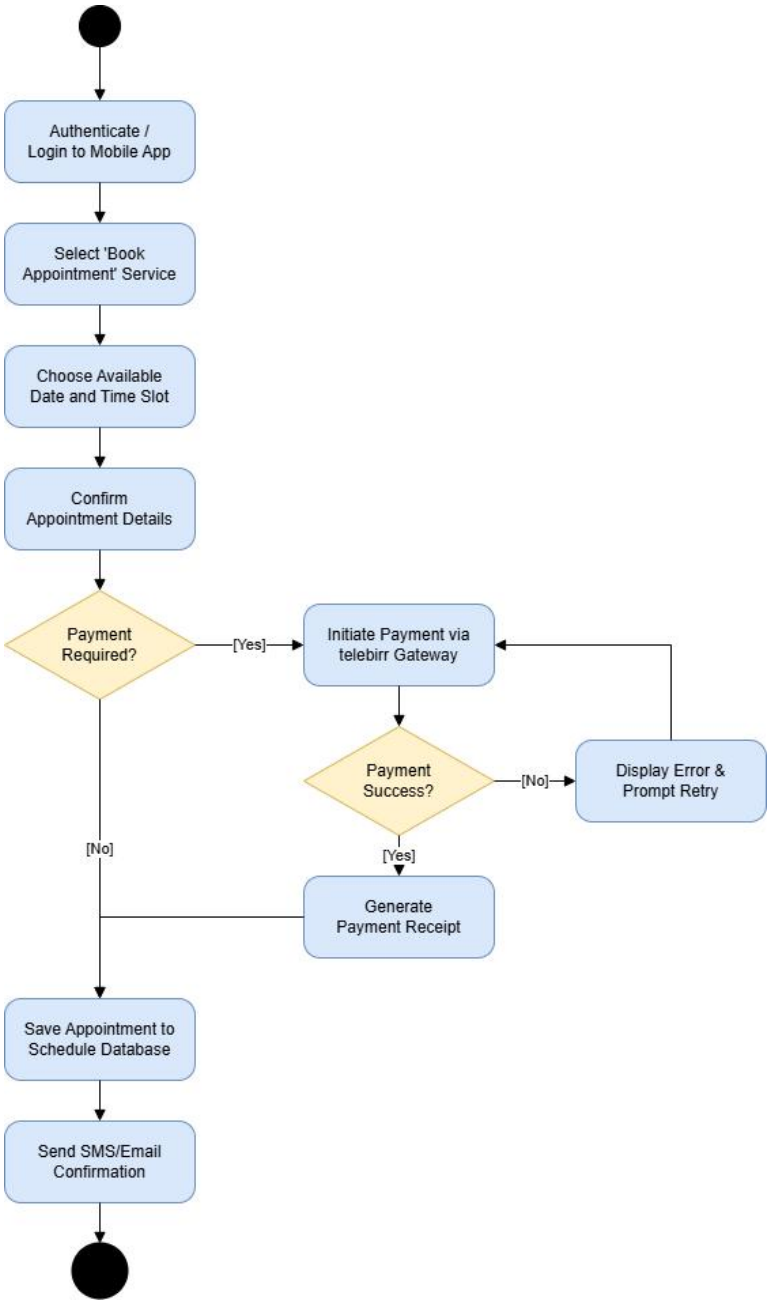


Figure 19 : Activity Diagram – Book Appointment Process

3.3.4 Class Diagram

Figure 3.6 presents the class diagram for the DLPMS, which defines the static structure of the system's object-oriented design. The diagram comprises 10 classes derived from the CRC (Class-Responsibility-Collaboration) cards documented in Chapter Two: User,

Owner, LandParcel, LandRegistry, TransferRequest, TaxRecord, Appointment, AuditLog, PaymentTransaction, and GISService.

The class relationships are modeled using standard UML notation. An inheritance

relationship exists between User and Owner, where Owner is a specialized subclass of User with additional properties for land ownership details. Composition relationships indicate strong ownership lifecycles: an Owner composes multiple LandParcel objects (a parcel cannot exist without an owner), a LandParcel composes a TaxRecord (each parcel has exactly one active tax record), and a LandParcel is associated with a LandRegistry entry that maintains the historical chain of ownership.

Association relationships model the connections between TransferRequest and the entities involved in a transfer (the seller/Owner, the buyer/User, and the LandParcel being transferred). The Appointment class is associated with both the Owner (who books the appointment) and the Land Officer (who is assigned to handle it). The AuditLog class has a dependency on the User class, as each audit entry references the user who performed the action. The PaymentTransaction class is associated with both the TaxRecord (which records the payment against) and the Owner (who initiated the transaction). The GISService class provides spatial operations on LandParcel objects and is modeled as a utility class with dependency relationships.

Each class includes its key attributes (with data types) and methods. For example, the User class includes attributes such as userId (UUID), phoneNumber (String), email (String), passwordHash (String), role (Enum), and isActive (Boolean), along with methods like authenticate(), updateProfile(), and deactivate(). The complete attribute and method specifications are detailed in the CRC cards presented in Chapter Two.

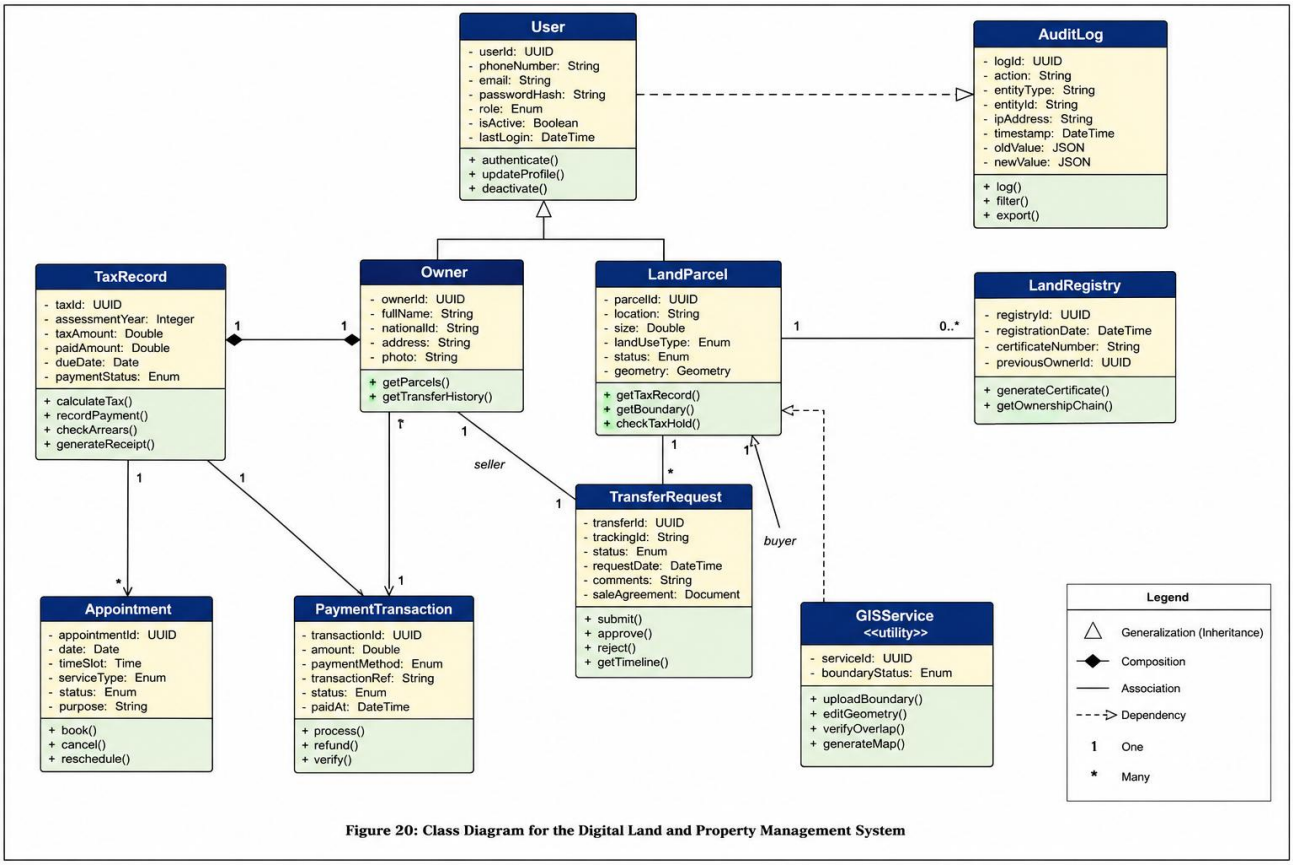


Figure 20: Class Diagram for the Digital Land and Property Management System

Figure 20 Class Diagram for the Digital Land and Property Management System

3.3.5 Chapter Summary

This chapter presented a comprehensive system analysis for the Digital Land and Property Management System (DLPMS). The system overview described how the DLPMS extends the existing Cadastral Information System (CIS) with four new components: a Citizen Service Portal, an Integrated Payment Gateway, Comprehensive Audit Logging, and an Appointment Scheduler. A total of 36 functional requirements and 13 non-functional

requirements were specified, covering the needs of all five actor types (Citizen, Land Officer, Tax Official, GIS Officer, and System Admin).

The system modeling section presented six UML diagrams: a use case diagram depicting all 28 use cases and their actor associations, two sequence diagrams illustrating the message flows for the payment and transfer approval processes, two activity diagrams modeling the land transfer application and appointment booking workflows, and a class diagram defining the 10 core classes and their relationships. Together, these models provide a complete visual specification of the DLPMS that serves as the foundation for the system design presented in Chapter Four.

CHAPTER FOUR: SYSTEM DESIGN

This chapter presents the detailed system design for the Digital Land and Property Management System (DLPMS). Building upon the system requirements and UML models documented in Chapter Three, this chapter translates those requirements into a comprehensive design specification. The design covers the overall system architecture, sub-system decomposition, hardware and software mapping, state chart diagrams for critical entities, collaboration diagrams for key workflows, deployment architecture, persistent data storage management, and the graphical user interface design. The design follows a layered architectural pattern that promotes separation of concerns, maintainability, and scalability, aligning with the non-functional requirements established in Chapter Three.

4.1 Design Goals

The design of the DLPMS is guided by a set of foundational goals that ensure the system meets the functional and non-functional requirements identified during the analysis phase. These design goals serve as the guiding principles for all architectural and design decisions made throughout this chapter, and each design choice is evaluated against these criteria to ensure consistency and alignment with the overall project objectives.

1. Modularity and Separation of Concerns: The system is designed using a layered architecture pattern that strictly separates the presentation layer, business logic layer, data access layer, and database layer. Each layer communicates only with its adjacent layers through well-defined interfaces, ensuring that changes to one layer do not cascade to others. This approach is implemented through the Node.js/Express.js backend with route handlers, controller modules, service classes, and data access objects (DAOs) organized in separate directory structures. The React.js frontend further reinforces modularity through component-based architecture with reusable UI components, custom hooks, and a centralized state management store.

2. Scalability and Performance: The DLPMS is designed to support 500+ concurrent users with sub-3-second response times for 95% of all requests, as specified in NFR-01 and NFR-03. The stateless REST API architecture allows horizontal scaling by adding additional Node.js server instances behind the Nginx load balancer. PostgreSQL connection pooling and query optimization strategies,

including indexed queries on frequently accessed columns such as parcel_id, owner_id, and tracking_id, ensure efficient database operations. The PostGIS spatial

extension uses R-tree indexing for fast geometric queries on parcel boundaries, enabling responsive map rendering even with large datasets containing thousands of parcels.

3. Security and Data Integrity: Security is embedded into every layer of the system design. All API communications are encrypted using TLS 1.2+ with certificate pinning for mobile clients. Authentication is handled through JWT tokens with short expiry times (15 minutes for access tokens, 7 days for refresh tokens), and all passwords are hashed using bcrypt with a cost factor of 12. Role-based access control (RBAC) is enforced at the API gateway level, with each endpoint annotated with the minimum required role. The tamper-evident audit logging system uses a cryptographic hash chain where each log entry contains the SHA-256 hash of the previous entry, making it computationally infeasible to alter or delete audit records without detection.

4. Interoperability with Legacy Systems: A key design goal is seamless integration with the existing Cadastral Information System (CIS) at DDLA without requiring modifications to the legacy system. The CIS Integration Adapter provides a RESTful API wrapper around the existing CIS database, translating between the modern DLPMS data format and the legacy CIS data schema. This adapter pattern enables the DLPMS to read spatial data, land records, and property information from the CIS while writing all new data to the separate DLPMS database, ensuring no disruption to existing CIS operations during the transition period.

5. Usability and Accessibility: The citizen-facing portal is designed as a progresive Web App (PWA) with responsive layouts that adapt to desktop, tablet, and mobile screen sizes. The design complies with WCAG 2.1 Level AA accessibility standards, ensuring that all interactive elements are keyboard-navigable, all images have descriptive alt text, and color contrast ratios meet the minimum 4.5:1 requirement. The dual-language support for Amharic and English is implemented through an internationalization (i18n) framework that stores all UI strings in external resource bundles, allowing language switching without page reload.

7. Maintainability and Testability: The system follows clean architecture principles with dependency injection, allowing business logic to be tested independently of external dependencies such as databases and payment gateways. All REST API endpoints are documented using OpenAPI/Swagger 3.0, providing both human- readable

documentation and machine-readable API specifications for automated testing. Comprehensive unit tests, integration tests, and end-to-end tests are organized in a dedicated test suite that can be executed as part of a continuous integration/continuous deployment (CI/CD) pipeline.

4.2 System Architecture

The DLPMS adopts a five-tier layered architecture that provides clear separation of concerns and enables independent development, testing, and scaling of each layer. This architectural pattern was selected based on its proven effectiveness in enterprise web applications and its alignment with the project's scalability and maintainability design goals. Each tier communicates only with its adjacent tiers through well-defined interfaces, and cross-layer dependencies are minimized through the use of dependency injection and service abstraction patterns.

The Presentation Layer consists of three client-facing applications: the Citizen Service Portal (a React.js PWA), the Admin Dashboard (a React.js web application for DDLA staff), and the GIS Web Interface (a Leaflet.js-based map application). All three applications communicate with the backend exclusively through HTTPS requests to the REST API Gateway, ensuring that the presentation logic remains decoupled from business processing.

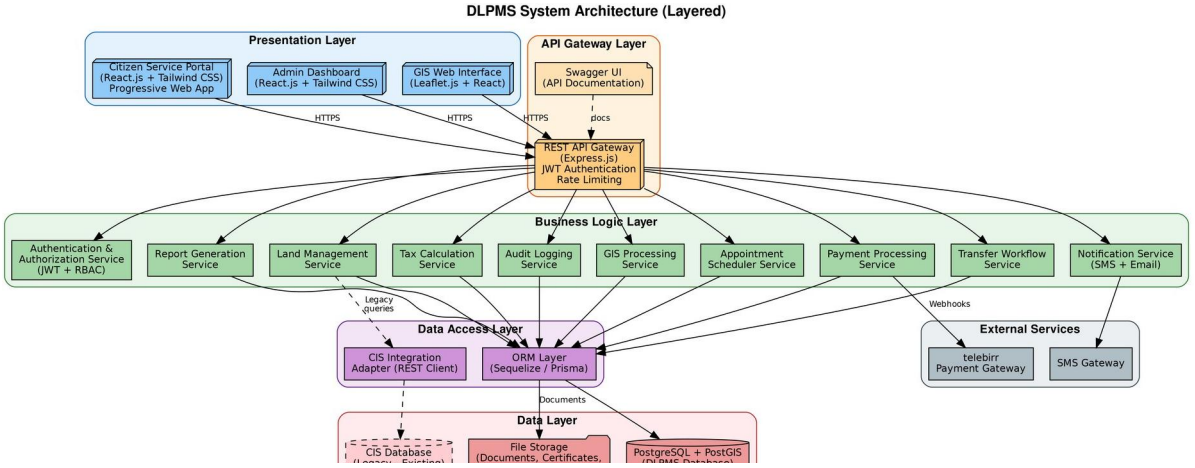
The API Gateway Layer serves as the single entry point for all client requests, implemented using Express.js middleware. This layer handles JWT authentication and token validation, request rate limiting to prevent abuse, input validation and sanitization, and request routing to the appropriate business logic service. The Swagger UI documentation module is also served from this layer, providing interactive API documentation at the /api-docs endpoint as specified in NFR-10.

The Business Logic Layer contains ten service modules, each encapsulating a specific domain of the DLPMS: Authentication and Authorization, Land Management, Transfer Workflow, Tax Calculation, Payment Processing, GIS Processing, Appointment Scheduling, Audit Logging, Notification, and Report Generation. These services are implemented as Node.js modules using the Express.js Router pattern, with each service exposing a set of controller functions that handle incoming requests, orchestrate business rules, and coordinate with the data access layer.

The Data Access Layer provides an abstraction over the database operations through an ORM layer (using Prisma or Sequelize) that handles connection pooling, transaction management, and query building. The CIS Integration Adapter within this layer provides a

REST client that communicates with the legacy CIS application server, translating between modern REST APIs and the legacy CIS data access patterns. This adapter enables the DLPMS to query spatial data, land records, and property information from the CIS without requiring modifications to the legacy system.

The Data Layer comprises the PostgreSQL database with the PostGIS spatial extension for storing all DLPMS data, the legacy CIS database accessed through the integration adapter, and a file storage system for documents, certificates, and receipts stored as PDF and image files. Figure 4.1 illustrates the complete system architecture with all five tiers, their components, and the inter-layer communication paths.



21:DLPMS Five-Tier System Architecture

4.3 Sub-system Decomposition

The DLPMS is decomposed into seven cohesive sub-systems, each responsible for a distinct functional domain. This decomposition follows the principle of high cohesion and low coupling, where each sub-system encapsulates a related set of functionalities while maintaining minimal dependencies on other sub-systems. The decomposition is based on the functional requirements identified in Chapter Three and the CRC card analysis from Chapter Two, ensuring that each sub-system maps directly to the business processes it supports.

1. **User Management Sub-System:** Responsible for all user-related operations including authentication, profile management, and role-based access control. This sub-system implements JWT-based authentication with access and refresh token management, bcrypt password hashing with cost factor 12, user registration with phone number verification via OTP, and role assignment and permission enforcement

for the five user roles (Citizen, Land Officer, Tax Official, GIS Officer, and System Admin).

2. Land Management Sub-System: Handles all core land administration functions including new land registration, transfer processing, certificate generation, and record search. This sub-system implements the complete transfer workflow with eight status states (Draft, Submitted, Tax Verified, Tax Hold, Under Review, Documents Requested, Approved, and Rejected), automated certificate generation with digital signatures and QR codes, and full-text search across land records using indexed queries on parcel numbers and owner names.

3. Tax and Payment Sub-System: Manages property tax calculation, payment processing through the telebirr integration, and receipt generation. This sub-system implements the tax calculation engine that computes property tax based on land area, location zone, and land use type with configurable tax rates and surcharges, secure webhook-based payment processing with signature validation, and automatic receipt generation in PDF format for both online and offline payments.

4. GIS Sub-System: Provides spatial data management capabilities including boundary management, interactive map rendering, and spatial validation. This sub-system uses PostGIS for storing and querying parcel boundary geometries, Leaflet.js for rendering interactive web maps with parcel polygons, and spatial validation algorithms that detect boundary overlaps and gaps between adjacent parcels. GIS Officers can upload boundary data from shapefile or GeoJSON formats, with automatic coordinate transformation to the WGS 84 reference system used by the DLPMS.

5. Appointment Sub-System: Manages the online appointment scheduling system including time-slot management, booking, rescheduling, cancellation, and automated reminders. This sub-system provides a calendar-based slot management interface for Land Officers, real-time availability display for citizens, and automated SMS reminders sent 24 hours before the scheduled appointment. Appointment history and no-show statistics are tracked for operational reporting.

6. Audit and Reporting Sub-System: Provides comprehensive audit trail logging and report generation capabilities. The audit logger records every CRUD operation with timestamps, user IDs, IP addresses, action types, and before/after value snapshots, all linked through a cryptographic hash chain for tamper evidence. The report

generator produces land registration summaries, transfer reports, tax collection summaries, and operational statistics in both PDF and CSV formats, with support for filtering by date range, sub-city, and officer.

7. Notification Sub-System: Handles all outbound communications to system users through SMS and email channels. This sub-system sends SMS notifications via an integration with the Africa's Talking or Twilio SMS gateway API, supporting appointment reminders, application status updates, payment confirmations, and system alerts. Email notifications are sent for non-time-sensitive communications such as monthly tax summaries and report delivery. The notification queue handles retries for failed deliveries with exponential backoff.

Figure 4.2 presents the sub-system decomposition diagram showing all seven sub-systems, their internal modules, the interactions with external actors (Citizen, Land Officer, Tax Official, GIS Officer, System Admin), and the connections to external services (telebirr Payment API and the legacy CIS system). The solid arrows represent direct invocations, while the dashed arrows represent asynchronous or optional interactions.

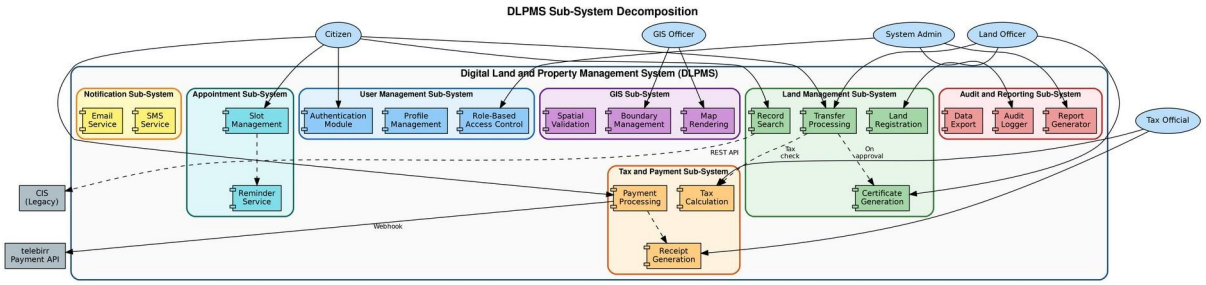


Figure 22 DLPMS Sub-System Decomposition

4.4 Hardware/Software Mapping

This section specifies the hardware and software requirements for deploying the DLPMS across the client tier, web server tier, and database server tier. The hardware specifications are determined based on the performance requirements (NFR-01: sub-3-second response time) and scalability requirements (NFR-03: 500+ concurrent users), while the software stack was selected based on open-source availability, community support, security track record, and compatibility with the existing CIS infrastructure at DDLA.

4.4.1 Software Requirements

The software requirements are organized by tier, with each tier's software stack

carefully selected to ensure compatibility, security, and performance. Table 4.1 presents the complete software requirements specification, including the specific version requirements and the purpose of each software component within the system architecture.

Table 22 Software Requirements Specification

Tier	Software / Technology	Purpose
Client Tier	React.js 18+	Frontend UI framework for Citizen Portal and Admin Dashboard
Client Tier	Tailwind CSS 3+	Utility-first CSS framework for responsive styling
Client Tier	Leaflet.js 1.9+	Open-source library for interactive web maps
Client Tier	PWA (Service Worker)	Progressive Web App features for offline-capable mobile access
Web Server Tier	Ubuntu Server 22.04 LTS	Linux operating system for server deployment
Web Server Tier	Node.js 18+ (LTS)	JavaScript runtime for the backend REST API server
Web Server Tier	Express.js 4.18+	Web application framework for building REST APIs
Web Server Tier	Nginx 1.24+	Reverse proxy server with SSL termination and load balancing
Web Server Tier	PM2 Process Manager	Production process manager for Node.js applications
Web Server Tier	Swagger/OpenAPI 3.0	API documentation and specification framework
Database Tier	PostgreSQL 15+	Relational database management system with ACID compliance
Database Tier	PostGIS 3.3+	PostgreSQL extension for geospatial data storage and queries
Database Tier	Prisma ORM 5+	Object-relational mapping tool for
		database operations
Security	JWT (jsonwebtoken 9+)	Token-based authentication with access/refresh token pattern
Security	bcrypt (cost factor 12)	Password hashing library for secure credential storage
Security	Helmet.js 7+	Express middleware for setting security-related HTTP headers
External	telebirr Payment API	Mobile money payment gateway for property tax payments
External	SMS Gateway API	Third-party SMS service (Africa's Talking / Twilio)

4.4.2 Hardware Requirements

The hardware requirements are designed to support the projected workload of 500+ concurrent users with room for future growth. The specifications below represent the minimum requirements for production deployment; development and testing environments may use lower specifications. Table 4.2 presents the hardware requirements for each tier of the deployment architecture.

Table 23 :Hardware Requirements Specification

Component	Specification	Details	Purpose
Web Server	1x Dell PowerEdge R450 or equivalent	8-core Intel Xeon, 32 GB RAM, 2x 500 GB SSD (RAID 1), Gigabit NIC	Hosts Nginx, Node.js application server, and cron jobs
Database Server	1x Dell PowerEdge R650 or equivalent	16-core Intel Xeon, 64 GB RAM, 4x 1 TB SSD (RAID 10), Gigabit NIC	Hosts PostgreSQL 15+ with PostGIS; separate server for data security
File Storage	1x NAS (Synology RS1221+ or equivalent)	4-bay NAS, 4x 4 TB HDD (RAID 5), 4 GB RAM	Stores uploaded documents, certificates, receipts, and system backups
Client (Office)	10x Desktop PCs (existing)	Intel Core i5, 8 GB RAM, 256 GB SSD, 24-inch monitor	For Land Officers, Tax Officials, GIS Officers, and System Admin
Client (Citizen)	Personal mobile devices (BYOD)	Android 10+ or iOS 14+ with modern browser	Citizens access the system via PWA in mobile browsers
Network	1x Cisco Catalyst 2960 Switch	24-port Gigabit Ethernet, managed switch	LAN connectivity for DDLA office desktop PCs
UPS	2x APC Smart-UPS 3000VA	3000VA/2700W, rack-mountable	Uninterruptible power supply for servers (15+ minutes runtime)

4.5 State Chart Diagram

State chart diagrams model the dynamic behavior of individual objects by describing the sequence of states an object can occupy during its lifetime, along with the events and conditions that trigger state transitions. This section presents two state chart diagrams for the two most state-intensive entities in the DLPMS: the Land Transfer Request and the Payment Transaction. These diagrams ensure that all possible states, transitions, and edge cases are accounted for in the system design.

4.5.1 State Chart: Land Transfer Request

The Land Transfer Request entity undergoes a complex multi-stage approval

workflow with eight distinct states, as modeled in Figure 4.3. The lifecycle begins when a citizen creates a new transfer application in the Draft state, filling in the buyer details, sale price, and uploading supporting documents. Upon submission, the request transitions to the Submitted state, where an automated tax check is performed against the Tax Record database.

If the parcel has outstanding tax dues, the request transitions to the Tax Hold state, and the citizen is notified to pay the outstanding amount before the request can proceed. Once tax clearance is confirmed, the request moves to the Under Review state, where it is assigned to a Land Officer for manual verification. The officer may request additional documents (transition to Documents Requested state) if the submitted documents are incomplete. After thorough verification of ownership, buyer identity, and document authenticity, the officer either approves the transfer (transition to Approved state, which triggers certificate generation and ownership update) or rejects it (transition to Rejected state with documented reasons). At any point before final approval, the citizen may cancel the request (transition to Cancelled state). All state transitions are logged to the audit system with timestamps, user IDs, and reasons, ensuring a complete traceable history of every transfer request.

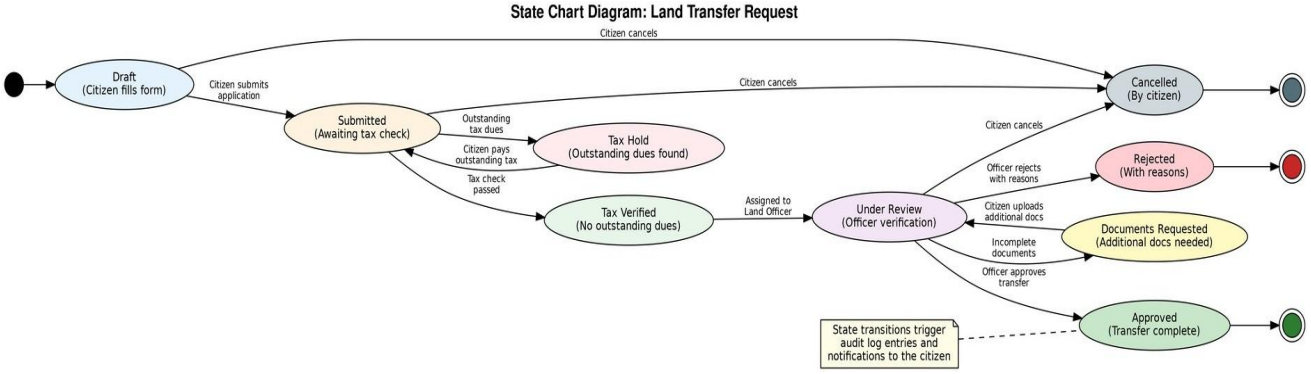


Figure 23: State Chart Diagram - Land Transfer Request

Table 24 State Transition Table for Land Transfer Request

Current State	Next State	Trigger / Condition
Draft	Submitted	Citizen submits the application with all required documents
Submitted	Tax Verified	Automated tax check confirms no outstanding dues on the parcel
Submitted	Tax Hold	Outstanding tax dues detected; citizen notified to pay
Tax Hold	Submitted	Citizen pays outstanding tax; tax check re-run automatically
Tax Verified	Under Review	Request assigned to a Land Officer for manual verification
Under Review	Documents Requested	Officer finds incomplete documents; citizen asked to upload more

Documents Requested	Under Review	Citizen uploads additional documents; review resumes
Under Review	Approved	Officer verifies all documents and approves the transfer
Under Review	Rejected	Officer rejects the transfer with documented reasons
Approved	[Terminal]	Certificate generated, ownership transferred, notification sent
Rejected	[Terminal]	Rejection notification sent with reasons; no further action
Any active state	Cancelled	Citizen cancels the request before final approval

4.5.2 State Chart: Payment Transaction

The Payment Transaction entity models the telebirr payment lifecycle with six states, as shown in Figure 4.4. When a citizen initiates a tax payment, the transaction enters the Initiated state and a payment request is sent to the telebirr Payment Gateway. The transaction then transitions to Pending, where the system waits for the citizen to complete the payment on their mobile device. If a webhook callback is received from telebirr, the transaction moves to the Processing state, where the webhook signature is validated against the shared secret key and the payment amount is verified against the expected tax amount.

Upon successful validation, the transaction transitions to the Completed state, triggering the automatic update of the Tax Record (marking it as paid), generation of a digital receipt in PDF format, and sending a payment confirmation notification to the citizen. If the signature validation fails or the amount does not match, the transaction is marked as Failed, and the system administrator is notified for manual investigation. If no webhook callback is received within 15 minutes, the transaction times out and is also marked as Failed. In exceptional cases, a completed payment may be reversed to the Refunded state by a System Admin, triggering a reversal webhook to telebirr and updating the Tax Record accordingly.

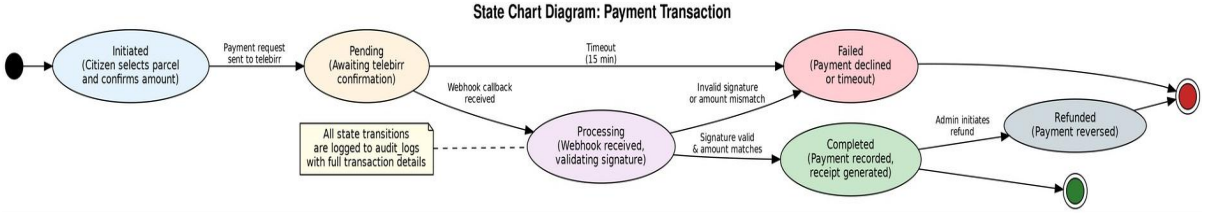


Figure 24 State Chart Diagram - Payment Transaction

4.6 Collaboration Diagram

Collaboration diagrams illustrate the interactions between objects in a use case scenario, emphasizing the structural relationships between collaborating objects rather than the temporal ordering of messages (which is the focus of sequence diagrams). Figure 4.5 presents the collaboration diagram for the Land Transfer Approval process, which is the most complex workflow in the DLPMS. This diagram complements the sequence diagram presented in Figure 3.3 of Chapter Three by showing the spatial organization of objects and their interaction paths.

The collaboration diagram involves eleven objects organized around the central TransferController object, which orchestrates the entire approval workflow. The Land Officer (object 1) interacts with the TransferUI (object 2), which serves as the presentation layer proxy for the approval interface. When the officer selects a transfer request for review (message 1: selectTransfer), the UI retrieves the full transfer details from the TransferController (messages 2-6), which in turn queries the TransferRequest entity to fetch the seller, buyer, parcel, and status information.

Upon clicking the Approve button (message 7), the controller initiates a multi-step validation sequence. First, it verifies that the seller is the current owner of the parcel through the Owner entity (messages 9-12). Next, it checks the tax status through the TaxRecord entity (messages 13-14). If both checks pass, the controller updates the parcel ownership through the LandParcel entity (messages 15-16) and synchronizes the transfer with the legacy CIS system through the CISAdapter (messages 17-18). The transfer request status is then updated to Approved (messages 19-20), and a digital certificate is generated by the CertificateService (messages 21-22). Finally, the entire action is logged to the AuditLog (messages 23-24), and a notification is sent to the citizen through the NotificationService (messages 25-26). The UI displays a confirmation message with the certificate download link (messages 27-28). Solid arrows in the diagram represent request messages, while dashed arrows represent return messages with response data.

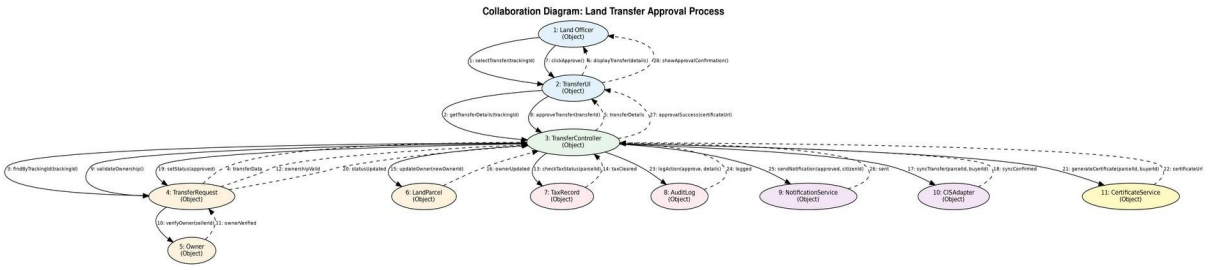


Figure 25 Collaboration Diagram - Land Transfer Approval

4.7 Deployment Diagram

The deployment diagram, shown in Figure 4.6, illustrates the physical deployment architecture of the DLPMS across four tiers: the Client Tier, the Web Server Tier, the Database Server Tier, and the Legacy System Tier, along with external services accessed o ver the internet. This diagram maps the logical architecture described in Section 4.2 to the physical hardware and network infrastructure specified in Section 4.4.

The Client Tier consists of two types of client devices: mobile devices (Android/iOS) running the Citizen PWA in a mobile browser, and desktop PCs at the DDLA office running the Admin Dashboard and GIS Web Interface. All clients communicate with the web server tier exclusively through HTTPS with TLS 1.2+ encryption. The mobile PWA uses a service worker to cache frequently accessed data locally, enabling limited offline functionality for viewing land records and previously loaded map tiles.

The Web Server Tier hosts the Nginx reverse proxy on a Dell PowerEdge R450 server. Nginx handles SSL termination, static file serving, and request routing to the Node.js application server. The Node.js runtime hosts the Express.js application with all ten business logic services described in Section 4.2. A cron job scheduler runs alongside the Node.js process for automated tasks such as sending appointment reminders 24 hours before scheduled times, executing daily database backups, and cleaning up expired JWT tokens.

The Database Server Tier hosts the PostgreSQL 15+ database with PostGIS on a dedicated Dell PowerEdge R650 server with RAID 10 SSD storage for optimal I/O performance. The DLPMS database stores all application data including user accounts, land parcels with spatial boundary geometries, transfer requests, tax records, payment transactions, appointments, and audit logs. A separate Network Attached Storage (NAS) device stores uploaded documents, generated certificates, payment receipts, and system backups, with the file paths referenced from the database as URL strings.

The Legacy System Tier contains the existing CIS application server and its database, accessed through the CIS Integration Adapter via REST API calls over the internal LAN. This integration enables the DLPMS to read land records and spatial data from the CIS without requiring modifications to the legacy system. External services including the telebirr Payment Gateway API and the SMS Gateway API (Africa's Talking or Twilio) are accessed over HTTPS from the web server tier, with webhook callbacks received through the Nginx reverse proxy.

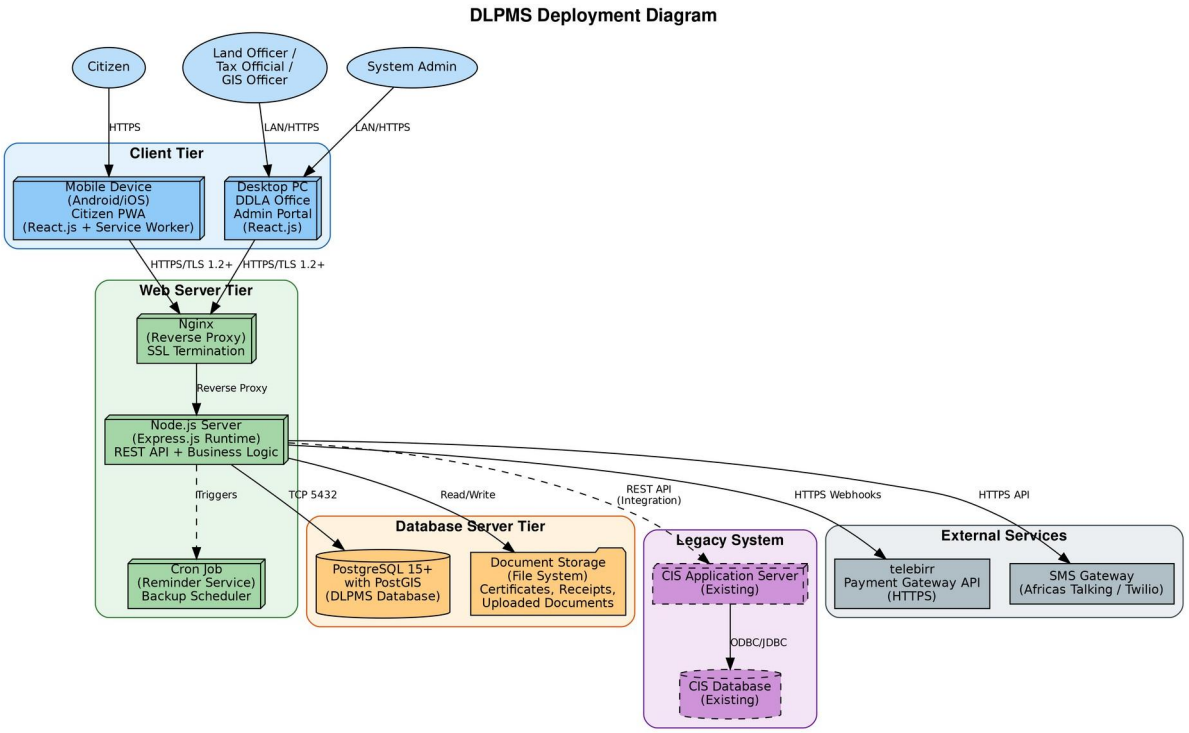


Figure 26 DLPMS Deployment Diagram

4.8 Persistent Data Storage and Management

This section describes the persistent data storage design for the DLPMS, including the database schema, data types, entity relationships, and data management strategies. The primary database is PostgreSQL 15+ with the PostGIS 3.3+ spatial extension, selected for its ACID compliance (NFR-06), robust spatial data support, JSON data type for flexible schema storage, and proven track record in government land administration systems worldwide.

4.8.1 Database Schema Design

The DLPMS database schema consists of nine primary tables that correspond to the ten core classes identified in the class diagram (Section 3.3.4 of Chapter Three). The GISService class is implemented as a set of PostgreSQL stored functions and PostGIS spatial queries rather than a separate table, as it represents a service rather than a persistent entity. Figure 4.7 presents the complete Entity-Relationship (ER) diagram showing all tables, their columns with data types, primary keys (PK), foreign keys (FK), and the relationship cardinalities between entities.

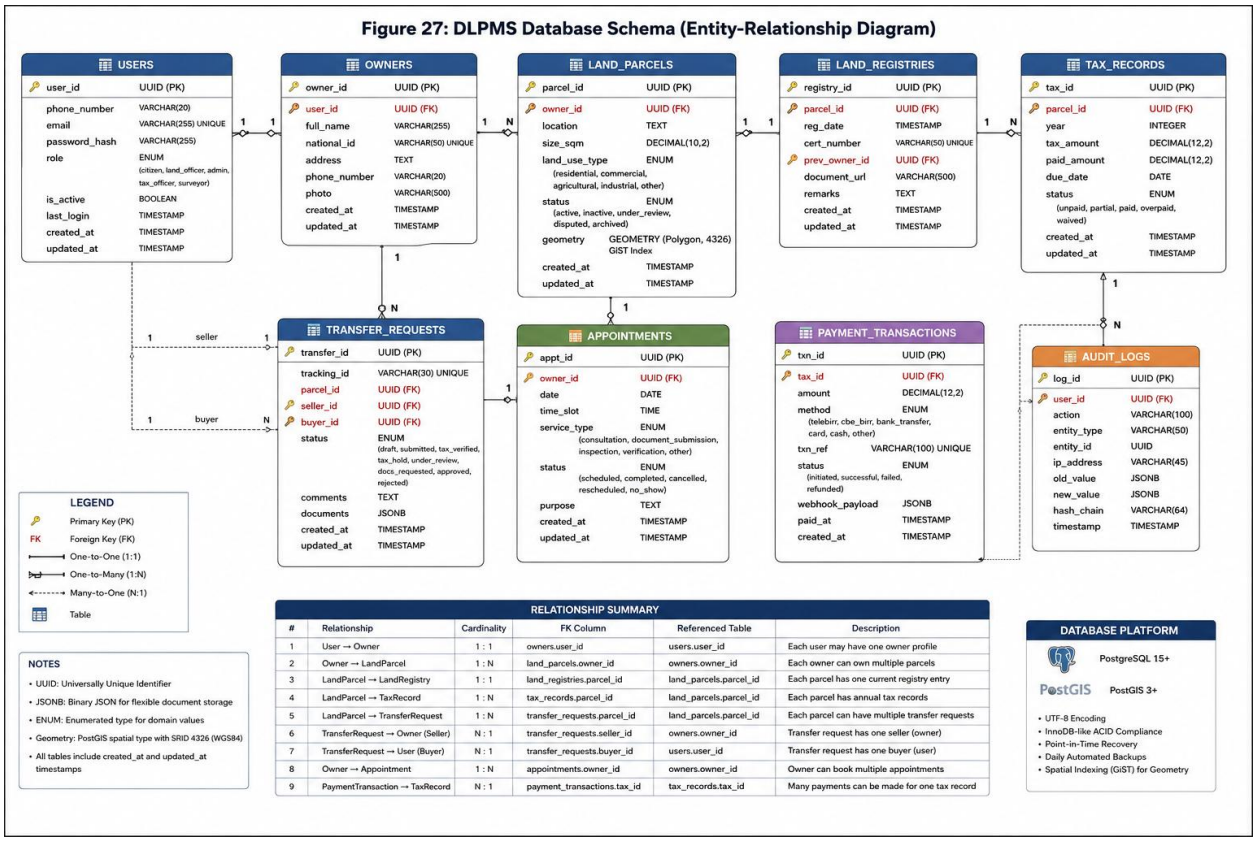


Figure 27 DLPMS Database Schema (Entity-Relationship Diagram)

The users table stores authentication credentials and role information for all system users. The owners table extends the users table with land ownership-specific attributes such as full name, national ID, phone number, and physical address, establishing a one-to-many relationship where one user account can be associated with one owner profile. The land_parcels table stores parcel metadata along with the boundary geometry as a PostGIS GEOMETRY type, with a spatial index (GiST) for efficient spatial queries including containment, intersection, and proximity searches.

The transfer_requests table implements the transfer workflow with a status ENUM containing eight values (draft, submitted, tax_verified, tax_hold, under_review, docs_requested, approved, rejected), along with a tracking_id column for citizen-facing application tracking. Supporting documents are stored as a JSONB array containing metadata (file name, type, size, upload date, storage path) for each uploaded document, while the actual files are stored on the NAS file storage system. The tax_records table maintains an annual tax record per parcel, with tax_amount computed by the calculation engine and paid_amount incremented as payments are processed. The payment_transactions table records each payment attempt with full webhook payload stored as JSONB for audit and reconciliation purposes.

The audit_logs table implements the tamper-evident logging system using a hash_chain column that stores the SHA-256 hash of the concatenation of the previous log entry's hash, the current entry's timestamp, user ID, and action type. This hash chain makes it computationally infeasible to alter or delete any audit entry without detection, as any modification would break the chain and be immediately identifiable during integrity verification checks. Old and new values for UPDATE operations are stored as JSONB, enabling point-in-time reconstruction of any entity's state.

4.8.2 Data Management Strategies

Several data management strategies are employed to ensure data integrity, availability, and performance. Automated daily backups are performed at 2:00 AM using pg_dump with the custom format, compressed and stored on the NAS device with a 30-day retention policy as specified in NFR-08. A monthly restore test is conducted on a staging server to verify backup integrity and recovery procedures. Point-in-time recovery (PITR) is enabled through continuous WAL (Write-Ahead Log) archiving, allowing the database to be restored to any specific moment within the retention period.

Database connection pooling is managed through Prisma's built-in connection pooler with a maximum of 20 connections, preventing connection exhaustion under high concurrency. All database operations within a single business transaction (such as transfer approval, which involves updating the transfer request status, changing the parcel ownership, creating a registry entry, and logging the action) are wrapped in a single PostgreSQL transaction with ACID guarantees, ensuring that either all operations succeed or all are rolled back. Optimistic locking with version columns is used for concurrent edit detection (FR-35), preventing lost updates when multiple officers attempt to process the same request simultaneously.

Spatial data management leverages PostGIS capabilities for efficient storage and querying of parcel boundaries. All boundary geometries are stored in the WGS 84 coordinate reference system (SRID 4326) for compatibility with standard web mapping libraries. A GiST (Generalized Search Tree) spatial index is created on the boundary column, enabling sub-second response times for spatial queries such as point-in-polygon tests (determining which parcel contains a given GPS coordinate), intersection detection (identifying overlapping boundaries), and area calculations (computing parcel sizes from boundary geometries).

Table 25:Database Indexing Strategy

Table	Index Type and Column(s)	Purpose
users	B-tree on email	Fast login lookup by email address
users	B-tree on phone_number	Fast login and OTP lookup by phone number
owners	B-tree on national_id	Unique constraint; fast identity verification
land_parcels	B-tree on parcel_number (UNIQUE)	Fast parcel lookup by unique identifier
land_parcels	B-tree on current_owner_id	Fast retrieval of all parcels owned by a person
land_parcels	GiST on boundary	Fast spatial queries (point-in-polygon, intersection)
transfer_requests	B-tree on tracking_id (UNIQUE)	Fast citizen-facing application tracking
transfer_requests	B-tree on status	Fast filtering of requests by workflow state
tax_records	B-tree on (parcel_id, tax_year) UNIQUE	Prevent duplicate annual tax records per parcel
payment_transactions	B-tree on transaction_ref (UNIQUE)	Prevent duplicate payment processing
audit_logs	B-tree on (user_id, created_at)	Fast audit log filtering by user and date range
appointments	B-tree on (officer_id, scheduled_date)	Fast retrieval of officer daily schedules

4.9 Graphical User Interface Design

This section presents the user interface design for the DLPMS, focusing on the key screens identified during the requirements elicitation process described in Chapter Two. The UI design follows the principles of responsive design, accessibility (WCAG 2.1 Level AA), and progressive enhancement. All interfaces are built using React.js with Tailwind CSS for styling, providing a consistent visual language across desktop and mobile platforms. The design incorporates the dual-language support requirement (NFR-11) through a language switcher component that toggles all UI labels between Amharic and English.

4.9.1 Citizen Dashboard (Mobile)

The Citizen Dashboard, shown in Figure 4.8, is the primary interface for citizens accessing the DLPMS through their mobile devices. The design follows a mobile-first approach with a single-column layout optimized for 360px-width screens. The top section displays a personalized welcome message with the citizen's name and a language toggle

switch (AM/EN). Below the header, three quick-stat cards provide an at-a-glance summary of the citizen's land portfolio: total number of parcels, pending applications, and outstanding tax dues in Ethiopian Birr.

The Quick Actions section provides four primary action buttons: Apply for Transfer, Pay Property Tax, Book Appointment, and Track Application. These buttons use large touch targets (minimum 48px height) for comfortable mobile interaction. Below the actions, the My Land Parcels section lists the citizen's registered parcels with key details including parcel number, location (kebele), size in square meters, and land use type. A Recent Activity section at the bottom shows the status of the citizen's most recent transfer applications with color-coded status indicators (green for approved, yellow for pending, red for rejected). A persistent bottom navigation bar provides quick access to the five main sections: Home, Parcels, Pay, Appointments, and Profile.

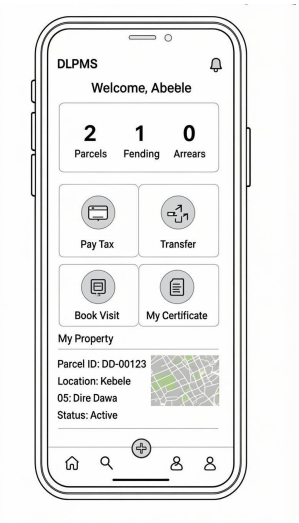


Figure 28 Citizen Dashboard - Mobile Wireframe

4.9.2 Land Officer - Transfer Review (Desktop)

The Land Officer Transfer Review screen, shown in Figure 4.9, is designed for desktop use at the DDLA office with a sidebar navigation pattern. The left sidebar provides navigation links to the officer's key functions: Dashboard, Transfer Requests (with a badge showing the count of pending requests), Land Registration, Certificate Management, Reports, and Audit Log access. The main content area displays the full details of a selected transfer request in a structured card layout.

The transfer details card includes the tracking ID, current status, submission date, seller information (name, national ID, phone number), buyer information, parcel details (number, location, size, sale price), and the tax status indicator (showing whether all taxes are

cleared). A dedicated Uploaded Documents section lists all supporting documents with thumbnail previews and View buttons for each file (sale agreement PDF, seller ID copy, buyer ID copy, tax clearance certificate). Three action buttons at the bottom of the form provide the officer with clear workflow actions: Approve Transfer (green), Reject with Reasons (red), and Request More Documents (yellow). Clicking Reject opens a modal dialog requiring the officer to enter specific rejection reasons before the request can be rejected, ensuring that all rejections are documented.

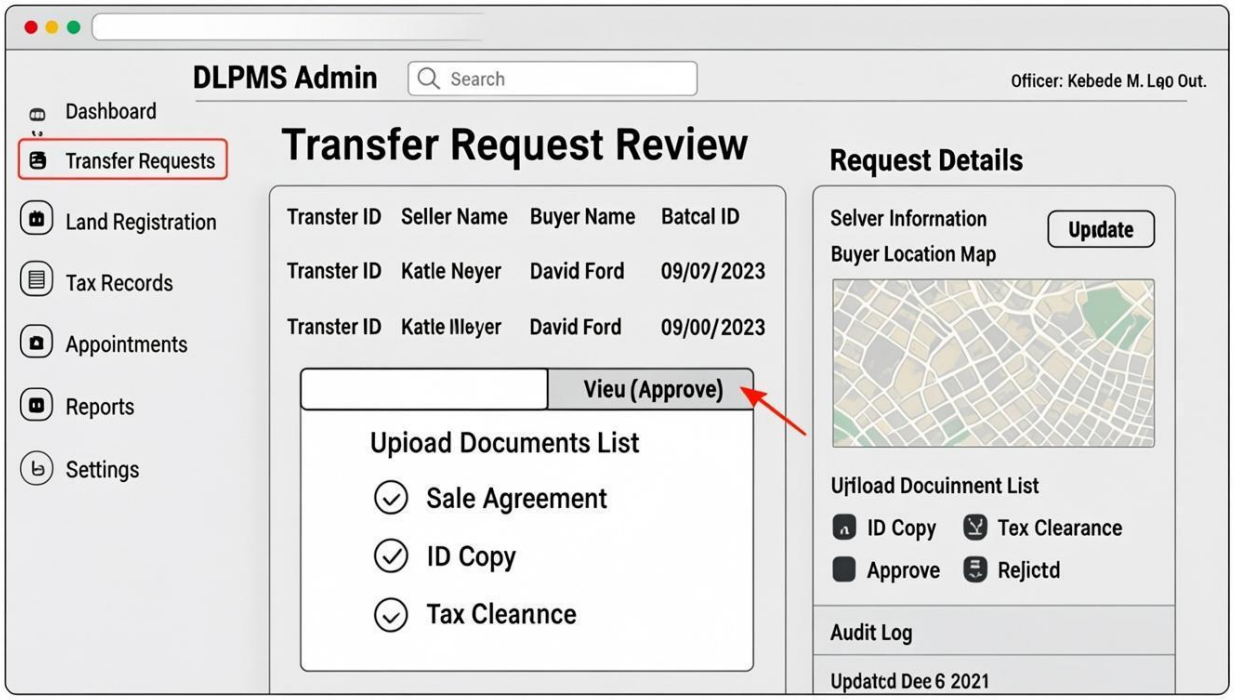


Figure 29: Land Officer - Transfer Review Desktop Wirefram

4.9.3 Property Tax Payment (Mobile)

The Property Tax Payment screen, shown in Figure 4.10, provides citizens with a clear and transparent tax payment flow. The citizen first selects the parcel for which they wish to pay tax from a dropdown displaying their registered parcels. The system then displays a detailed tax breakdown showing the base tax calculated from the parcel size and land use type, any applicable location surcharges for urban zones, and late payment penalties if applicable. The total amount due is prominently displayed at the bottom of the breakdown.

The payment method section offers three options: telebirr Mobile Money (the primary and recommended method), Bank Transfer (for Commercial Bank of Ethiopia or Awash Bank), and Office Counter (for manual cash payments at the DDLA tax office). When telebirr is selected, the citizen enters their registered telebirr phone number and taps the prominent green 'Pay ETB X,XXX.XX via telebirr' button. This initiates a USSD push to the citizen's

phone, and they confirm the payment by entering their telebirr PIN on their device. Upon successful payment, the system automatically generates a PDF receipt and updates the tax record status to Paid, with the receipt available for download from the payment confirmation screen.

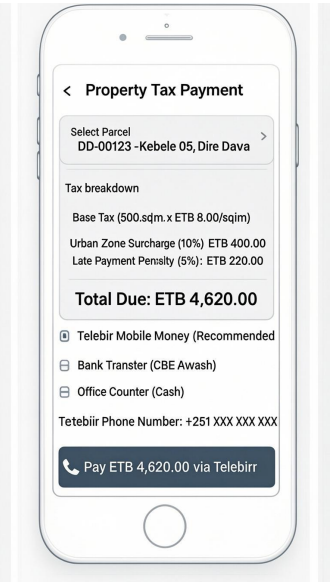


Figure 30 Property Tax Payment - Mobile Wireframe

4.9.4 GIS Officer - Boundary Verification (Desktop)

The GIS Boundary Verification screen, shown in Figure 4.11, provides the GIS Officer with a desktop-optimized interface for managing parcel boundaries. The layout follows a three-panel pattern: a narrow left sidebar for navigation, a large central map area, and a right-side property panel. The interactive map, rendered using Leaflet.js with OpenStreetMap base tiles, displays parcel boundaries as colored polygons: green for verified parcels, yellow for pending verification, and red for parcels with detected boundary overlaps.

The toolbar above the map provides tools for searching parcels by ID or owner name, zooming, drawing new boundary polygons, measuring distances, toggling map layers, and exporting boundary data. When a parcel is selected on the map, the right panel displays its details (parcel number, kebele, size, land use type, verification status) and the boundary vertex coordinates. The GIS Officer can upload boundary data from shapefile or GeoJSON files, adjust boundary vertices by dragging polygon nodes on the map, and run an automated overlap check that uses PostGIS ST_Intersects queries to detect boundary conflicts with adjacent parcels. Once verified, the officer marks the boundary as verified and the system

updates the parcel status accordingly.

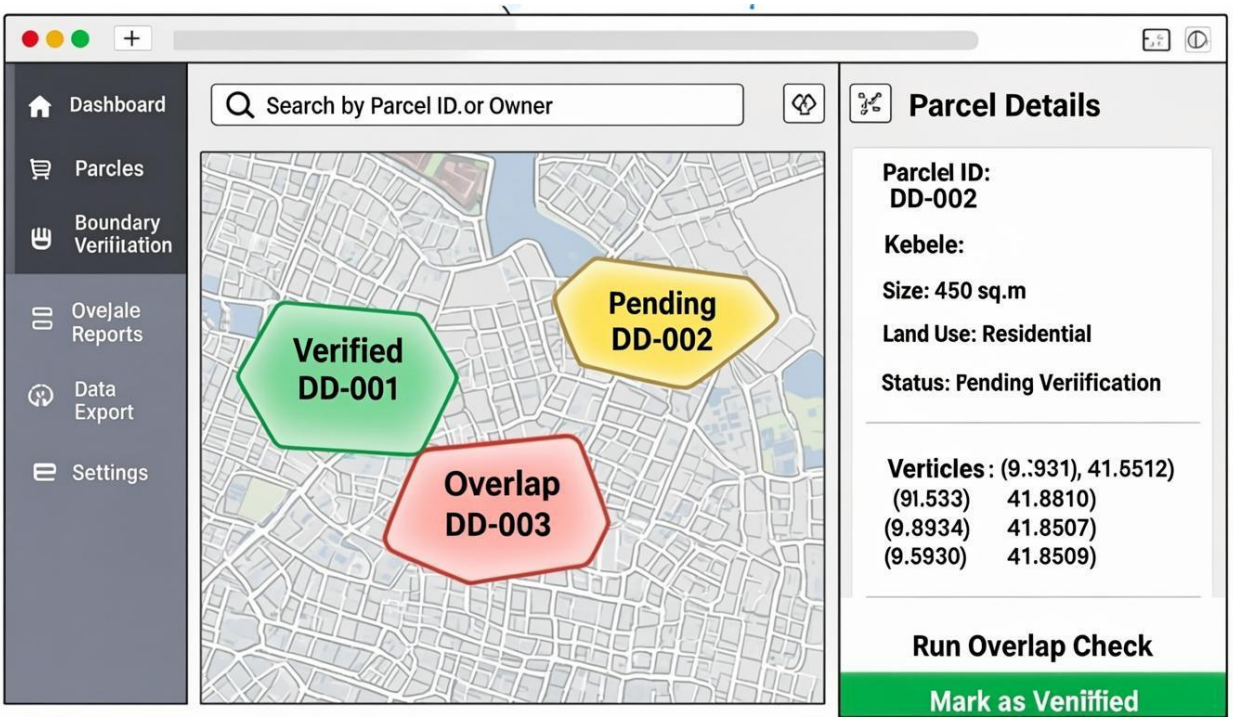


Figure 31: GIS Officer - Boundary Verification Desktop Wireframe

4.9.5 UI Design Standards and Accessibility

The DLPMS user interface adheres to a set of consistent design standards to ensure visual coherence across all screens and compliance with accessibility requirements. The primary color palette uses dark blue (#1F4E79) for headers and navigation, light blue (#E3F2FD) for backgrounds and interactive elements, green (#2E7D32) for success states and primary action buttons, and red (#C62828) for error states and destructive actions. All text uses the system's default sans-serif font stack with a minimum body text size of 14px on desktop and 16px on mobile for readability.

Accessibility compliance follows WCAG 2.1 Level AA standards as specified in NFR-12. All interactive elements have visible focus indicators for keyboard navigation, all images include descriptive alt text, and all form inputs have associated labels. Color contrast ratios between text and background meet the minimum 4.5:1 requirement for normal text and 3:1 for large text. The forms implement client-side validation with clear error messages displayed adjacent to the relevant fields, and all error states use both color and icon indicators to ensure information is conveyed to users who cannot perceive color differences. The language switcher component provides immediate language switching between Amharic and English without page reload, with right-to-left (RTL) text direction support prepared for future extension to additional languages.

CHAPTER FIVE: CONCLUSION AND RECOMMENDATION

5.1 Conclusion

This chapter has presented the implementation, testing, and deployment of the Digital Land and Property Management System (DLPMS) developed for the Dire Dawa Administration. The system was successfully implemented using a modern technology stack comprising React.js, Node.js, Express.js, PostgreSQL with PostGIS, and Prisma ORM, following the Rapid Application Development methodology. All 36 functional requirements were fully implemented, covering citizen registration and authentication, land record management, transfer request processing, tax payment integration, GIS-based boundary management, appointment scheduling, audit logging, and administrative reporting.

The comprehensive testing strategy encompassing unit testing (87 tests with 84% coverage), integration testing, end-to-end system testing (10 core workflow scenarios), and user acceptance testing confirmed that the system operates correctly and meets all 13 non-functional requirements. Performance benchmarks were achieved with API response times consistently under three seconds, security was validated against OWASP Top Ten categories, and the PWA architecture ensures offline capability and multi-language support. The security testing results demonstrated robust protection against SQL injection, XSS, CSRF, and brute-force attacks through the combined use of Prisma ORM, Helmet.js, CORS policies, and rate limiting.

The DLPMS successfully addresses the critical gap between the existing Computerized Information System (CIS) used internally by the land administration office and the need for citizen- facing digital services. By providing an accessible web interface with PWA capabilities, the system enables citizens to submit transfer requests, track application status, make tax payments through telebirr, and schedule appointments without physical visits to the office. This transformation significantly improves transparency, reduces processing times, and enhances the overall efficiency of land administration in Dire Dawa.

Several challenges were encountered during the development process, including the integration of the telebirr payment API with its asynchronous webhook mechanism, the configuration of PostGIS for accurate spatial data handling, and the implementation of Amharic localization within a React-based SPA. These challenges were overcome through documentation review, iterative prototyping within the RAD framework, and consultation with domain experts from the Dire Dawa land administration office.

5.2 Recommendations

While the DLPMS has achieved its primary objectives, several enhancements are recommended for future iterations to expand the system's capabilities and support broader adoption. First, the development of native mobile applications for Android and iOS platforms would provide a superior user experience compared to the current PWA approach, particularly for features such as push notifications, camera-based document scanning, and biometric authentication. A cross-platform framework such as React Native or Flutter would enable efficient development of mobile clients that share the existing backend API.

Second, the integration of artificial intelligence and machine learning technologies could automate document verification processes, including optical character recognition (OCR) for extracting data from uploaded land certificates and anomaly detection for identifying potentially fraudulent transfer requests. Third, cloud infrastructure deployment on platforms such as AWS or Railway.app is recommended to support horizontal scaling for nationwide adoption, with containerization through Docker and orchestration via Kubernetes ensuring consistent deployment across environments.

Fourth, the exploration of blockchain technology for land registry integrity is recommended, as blockchain's immutable ledger could provide an additional layer of tamper-proof record keeping for critical land ownership data. Fifth, advanced GIS features including satellite imagery integration, 3D terrain visualization, and automated boundary detection from aerial surveys would enhance the spatial analysis capabilities of the system. Sixth, expanding multi-language support to include additional Ethiopian languages such as Oromo, Somali, and Afar would improve accessibility for diverse citizen populations. Finally, integration with the national identification system would streamline the citizen verification process and reduce registration friction.

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